RPMGLOBAL

Mopani AMT Solution Design & Deployment Scope

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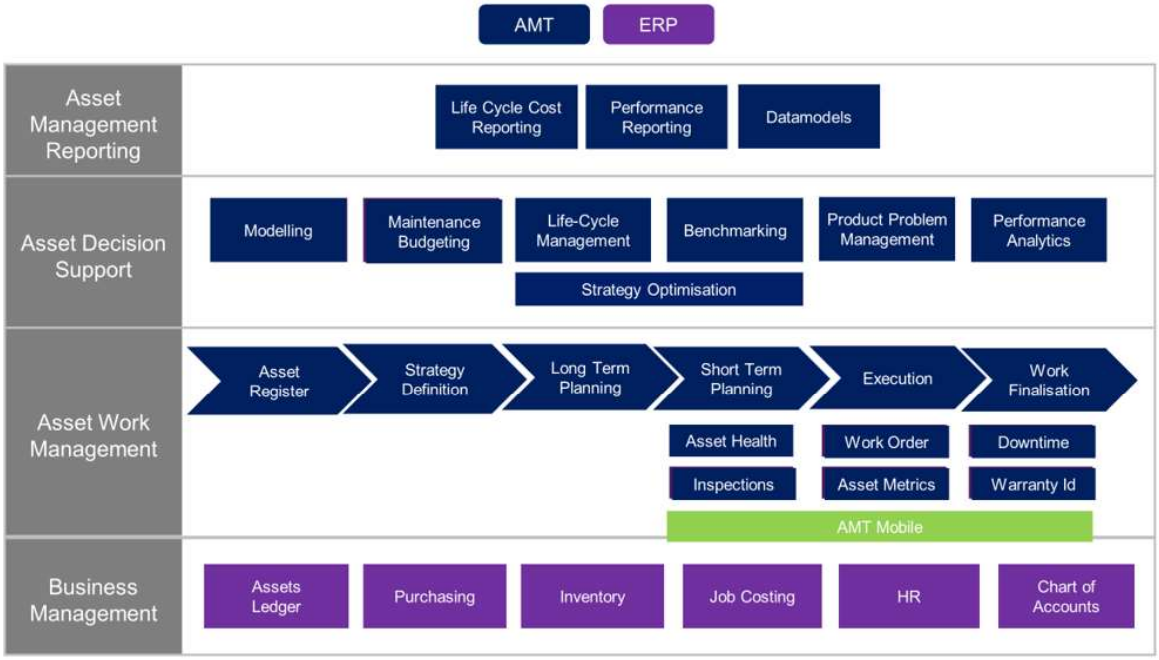
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1. AMT Footprint Design

Figure 1 AMT Footprint



AMT will be implemented in a two-phase approach.

Work Management: AMT integrated Database

Phase 1:

Work Management - July '24 Demo only for 4 Assets excluding integration.

Work Management on TMM3(Mobile assets) & Shafts by the end of October'24

Phase 1 Stage 2: Work Management including all assets and sites by end of January'25.

Phase 2: AMT Mobile + Integration on external systems defined in the integration scope by Mopani.

The solution design is for Phase 1 only.

FLOW CHART NOTE:

The purpose of each flow chart is to provide a high-level overview of the steps involved in a process and provide context for AMT functionality identified. The flowcharts do not replace the existing company process flow, and the business may integrate AMT processes into their business processes. A business process activity stream is undertaken by the customers' change management team in conjunction with the project team SME's with RPMGlobal support will determine the ultimate addition to or changes in business processes .

The corresponding AMT functionality to be implemented is listed in the table below:

Table 1 AMT Functionality To Be Deployed

AMT Functionality	Description
DLCC & Budgeting (Dynamic Life-Cycle Costing)	• Life-Cycle Costing and analytics • Maintenance Cost Budget • Advanced Equipment Selection Modelling • Intelligent Cost Analyst Service • Long Term Planning • Health-Condition Based Ruled Engine for AMT (Phase 2)
Equipment Management (EM) Issues	• Quantify product problems (financial cost, productivity cost) • Workflow for actioning product problem resolutions
Planning and Scheduling	• Backlog management • Work order planning • Work order scheduling • Equipment Status (Shift log) • Downtime Capture • Equipment Performance KPI reporting • Maintenance Efficiency KPI reporting
Asset Health	• Condition Based Maintenance
Asset Decision Support	• Maintenance Strategy Optimisation • Benchmarking • Asset Life Optimisation • Component Performance Analysis
Other Functionality	• Dashboard • AMT-IC custom report writing • Component Based Depreciation • Multi-currency

2. AMT Solution Processes

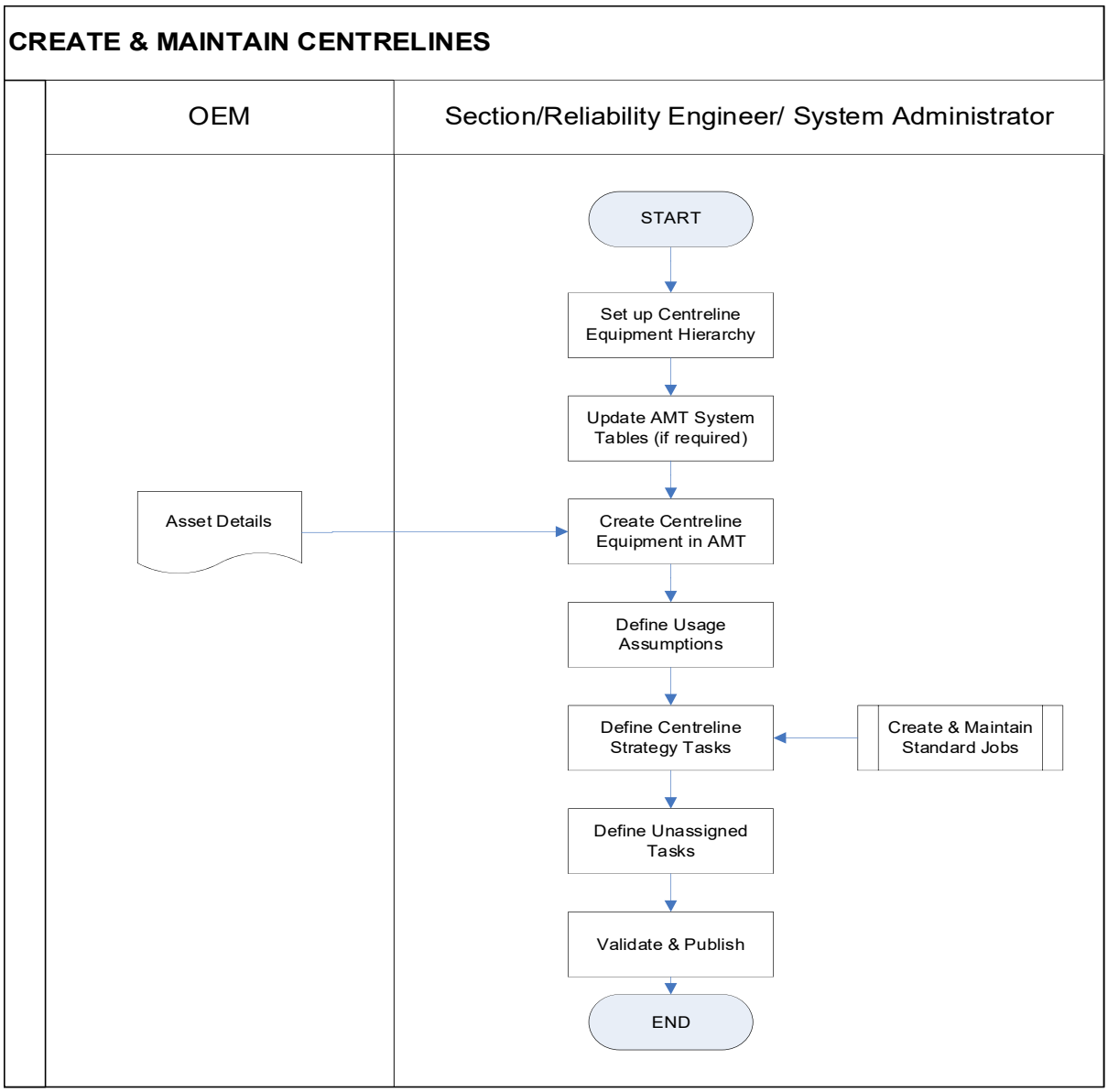
2.1. Create and Maintain Centrelines

A Centreline is a generic template for a type/model of equipment. The detail of a Centreline is primary its strategy tasks. Strategy Tasks are the basis for the equipment's life cycle costing (LCC) and drives forecasting. A centreline can be used as the basis for real equipment and for modelling equipment.

The objectives of the process are:

Objective	Importance/Benefit
To create and maintain equipment life cycle cost templates (i.e. centreline)	Used for Benchmarking and copy into Modelling and Real Equipment.

Figure 2 Centreline Process Flow Chart



Process steps performed using AMT:

Process Flow Step	AMT Function/Report	Notes
Update Centreline Equipment Hierarchy	Branch Administration Site Administration Fleet Administration	
Update AMT System Tables	System Tables > Maintenance	
Create Centreline Equipment in AMT	Create Blank Equipment - Centreline>Add New Centreline	
Define Usage Assumptions	Equipment Editor > Term Annual Utilisation	Usage Profile Manager only if Centrelines is configured for Advanced utilisation
Define Centreline Strategy Tasks	Equipment Editor > Tasks	

Process Flow Step	AMT Function/Report	Notes
Define AMT Unassigned Tasks	Equipment Editor > Tasks	Edit as required (First, Frequency, Cost)

The following outputs are expected from this process:

Output	Description
New Centreline Equipment	Available for copying to Modelling and Real Equipment
Full life-cycle cost projection for the equipment.	Generate a life cycle Projection Summary and Projection Details reports

Process Triggers:

Creation and/or update of a centreline or Equipment in ERP which is relevant to AMT. The process will be manually maintained in AMT.

Key Solution decisions regarding the **Centrelines** process:

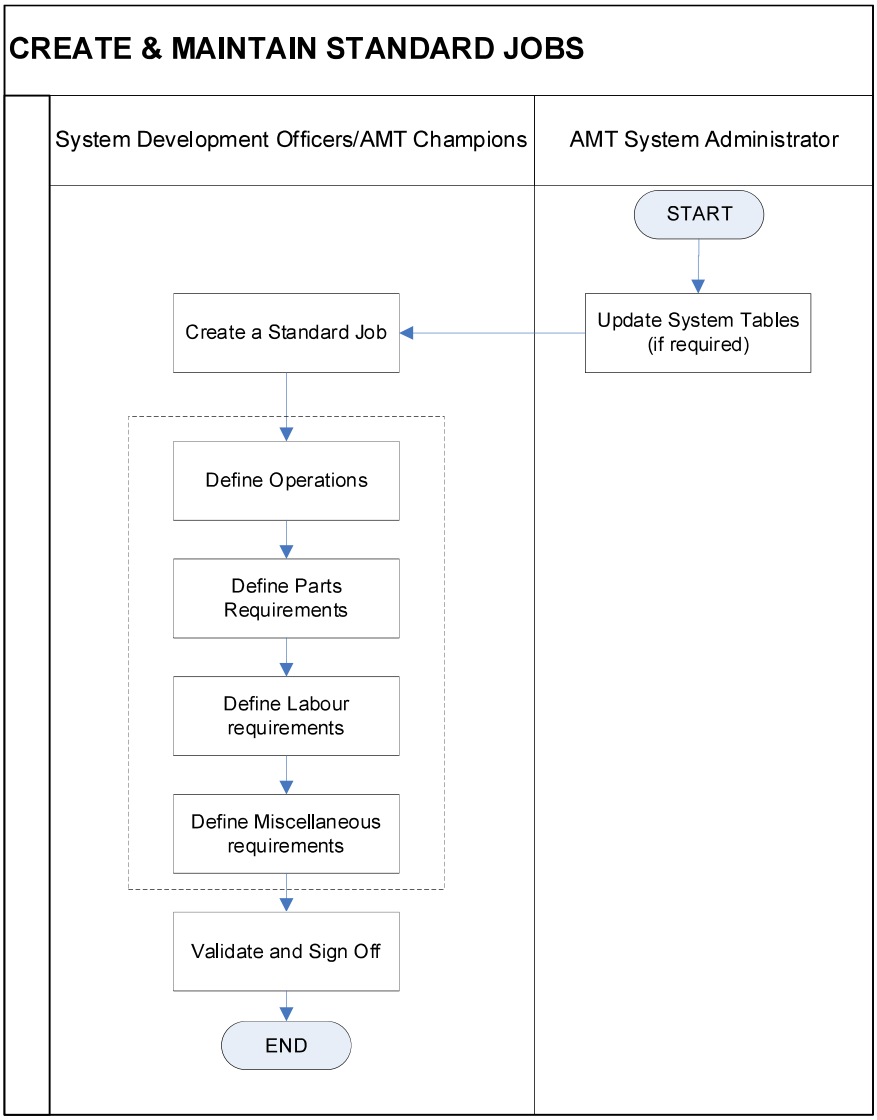
- AMT is the master for Centrelines.

2.2. Create and Maintain Standard Jobs

The objectives of the process are:

Objective	Importance/Benefit
To create and maintain standard jobs	If linked to strategy tasks the benefits of standard jobs are: • Improve cost forecast accuracy • Improve labour forecast accuracy • Enable piece part forecasting Can be imported into an ad-hoc work order to define the work order's: • Planned costs (work order's budget) • Planned duration • Planned parts, labour and miscellaneous requirements

Figure 3 Standard Jobs Process Flow Chart



Process steps performed using AMT:

Process Flow Step	AMT Function/Report	Notes
Update System Tables	System Tables > Parts & Standard Jobs System Tables > Organisation System Tables > Equipment System Tables > Maintenance	
Create a Standard Job	Standard Job Manager > Standard Job	
Define Operations	Standard Job Manager > Standard Job > Operations	
Define Parts requirements	Standard Job Manager > Standard Job > Parts	
Define Labour requirements	Standard Job Manager > Standard Job > Labour	
Define Miscellaneous requirements	Standard Job Manager > Standard Job > Misc	
Validate and Sign Off	Standard Job Manager > Standard Job	

The following outputs are expected from this process:

Output	Description
Standard Jobs	Library of standard jobs which can be • Linked to strategy tasks in Centreline, Modelling and/or Real equipment. • Accurate Pricing per customer. • Accurate Bills of Material that can be used to forecast parts for real assets. • Imported into ad-hoc work orders to define the work order's planned scope, resources, and costs.

Process Triggers:

A need for a new standard job for an existing or new centreline or equipment in AMT, the process will be manually maintained in AMT.

Update of a centreline or Equipment, in ERP which is relevant to AMT.

Key Solution decisions regarding the **Standard Job** process:

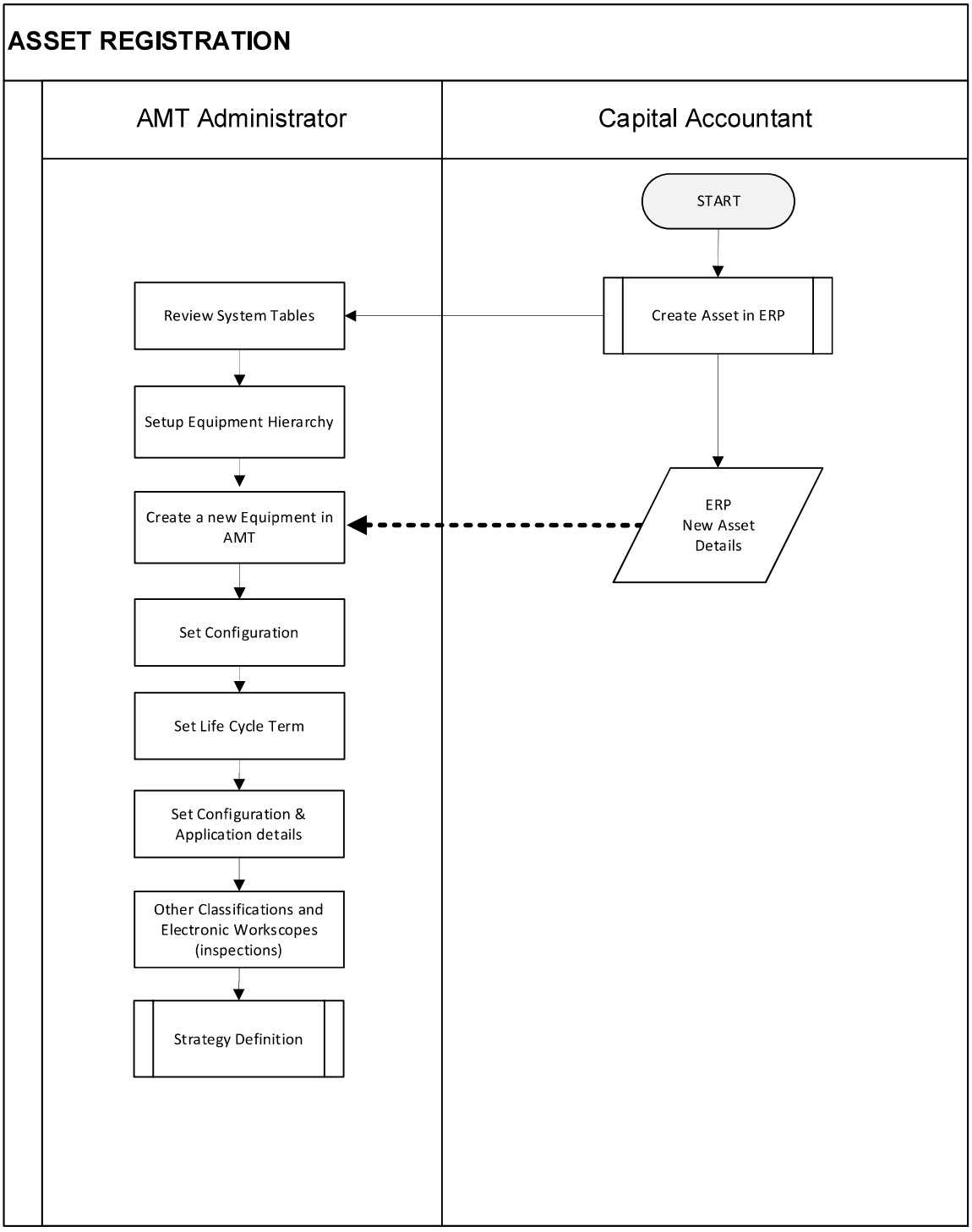
- ERP is the master for parts and parts pricing.
- All parts master data inventory in the ERP will be integrated to AMT through Part, Parts Prices & Component Inventory Parts interface.
- Only one Parts Pricing Group is required for AMT, which will be called "Default".
- Source of Supply will be mapped to "ERP".
- Part Number will be the Mopani's inventory's part number.
- Labour costing will not be done in AMT and as such the Labour Rates will not be defined in AMT.
- Miscellaneous category and Cost element (GL code) will be used to distinguish different non-Parts item such as contractor equipment, service cost from third party etc.
- The AMT work order, by design, does not hold Planned expense elements at the item level (i.e. per part line, labour line, misc. line). AMT only holds **actual** expense elements at the item level, updated by EPF UpdateWorkOrder or Classic Work Order Details interfaces.

2.3. Asset Registration

Asset registration is the process of registering a maintainable asset in AMT. The objectives of the process are:

Objective	Importance/Benefit
Register an asset in AMT	Enables AMT to generate a LCC for the equipment and analyse past and future performance. Enable users to plan and schedule work orders in AMT. Enable AMT to calculate the life-cycle cost (LCC) of the asset
Register asset Details in AMT	Create an asset in AMT using information from ERP. This is a critical step as it aligns assets between both systems, which is required for integration.

Figure 4 Asset Registration Process Flow Chart



Process steps performed using AMT:

Process Flow Step	AMT Function/Report	Notes
Review & Update AMT System Tables	System Tables > Organisation System Tables > Finance System Tables > Equipment System Tables > Usages & Measurements	Only check the system tables that are relevant to the new asset.
Create Fleet AMT Fleet	Fleet Administration	Only perform this step if a new Fleet is required. Enable availability forecasting for Mine maintenance fleets and set availability parameters.
Create a New Equipment in AMT	Create Blank equipment - Real	
Create & Link Usage Profile	Equipment Editor > Base Details > Utilisation Usage Profile Manager Equipment Editor > Term	This step can only be performed once the new equipment exists in AMT. Only applicable for equipment with more than one UOM's Most equipment will be configured as Simple Utilisation.
Enable "Auto Update" utilisation	Equipment Editor > Term	Only applicable to equipment configured with "Simple utilisation" mode.
Update Configuration & Application Details	Equipment Editor > Configuration & Application	Only complete if required.
Update Classifications	Equipment Editor > Classifications	Relevant fields: Category, Cost Centre, Cost Bearer
Update Scheduled Hours and Availability Targets	Scheduled Hours	Only required for assets that will report availability in AMT. Scheduled hours are required for reporting Physical Availability.
Set Quick Watch Parameters	Multi-Editor Quick Watch screen	• No Recent Usage Reading • Overdue Component Changeouts • No Usage Reading Tol. Days • Non-Strat. WO "High Cost" • Non-Strat. WO "High Cost" Min. $ • WO Cost Var. $ • WO Cost Var. % • WO Cost Variance
Sign Off	Equipment Editor > Base Details Equipment Editor > Term Equipment Editor > Classifications Equipment Editor > Configuration & Application	Review all the asset details are accurate as per the equipment change and commissioning control document.

Outputs from this process are:

Output	Description
Updated AMT Asset Hierarchy.	MOPANI's asset hierarchy is visible and up to date in AMT.
New assets are registered in AMT.	Asset is ready to be managed in AMT.

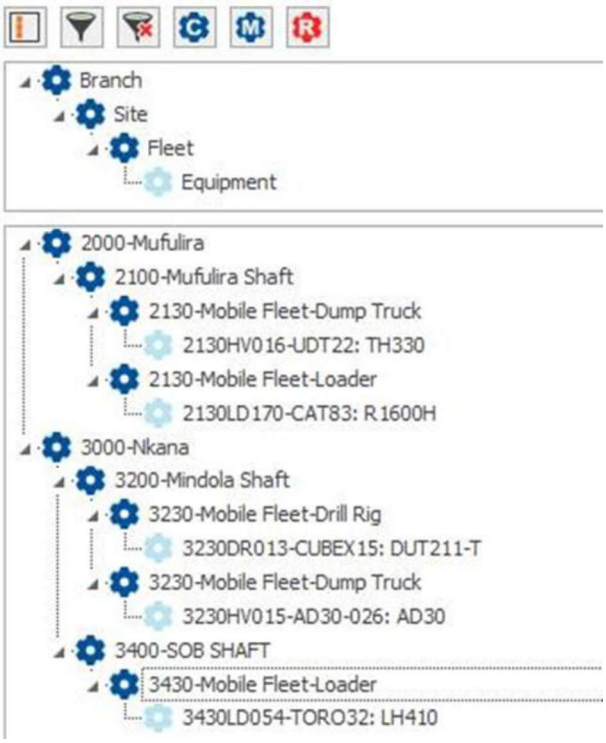
Process Triggers:

A new equipment needs to be added in AMT, the process will be manually managed in AMT.

All assets registered in AMT from the ERP will be real assets in AMT.

Key Solution decisions regarding the **Asset Registration** process:

- ERP is the master for all Mobile and Fixed plant assets.
- Mopani will have multiple Branches with a prefix of a location code from the ERP system i.e. 3000-Nkana & 2000-Mufulira, A site will be associated with a respective branch that it belongs to and represented with a prefix site code i.e. 3100-Mindola North Shaft site belonging to Branch 3000-Nkana. The Branch, Contractor will replace the old MCM Branch where contractor equipment was registered.
- Fleet will be grouping assets that belong to a certain fleet type i.e. 2130-Mobile Fleet-Dump truck.
- Fleet will be used to group equipment under the same functional location or processing area such as K160 and CRU001.
- Scheduled Hours and Availability Target parameters per asset, required for reporting availability, are maintained in AMT.
- The life-cycle term of an asset will be maintained in AMT.
- Usage Profiles, if required, are maintained in AMT. Usage Profiles are linked to equipment in AMT, we will be using current projections during initial setup.
- The alternative projections will be used for "What If" scenarios.
- Budget projections will be used for refurbished and new assets that Mopani will procure in the future to track their life cycle cost variances.



- All equipment will be associated with cost centre as provided from ERP and AMT equipment systems table
- The logging of asset commissioning, decommissioning, transfers and termination documents will be used to track all changes of the assets in AMT.
- Resource scheduling will be at a work group level for all sites.
- Mopani will be using simple planning (this is where Downtime is not captured in AMT) and advance planning (this is where Downtime is captured in AMT). Fixed plant is recommended to use simple planning but where fixed assets are flagged as critical by Mopani together with mobile equipment will use advance planning.
- The registration counter will be "1" for all assets that are registered in the beginning of the project, then when an asset starts another life, it will then change to registration counter "2".
- The equipment can be set by date or usage, the start and commission date can be same if original dates are unknown. The annual utilisation for Mobile Assets will be set to 5000 if Hours are not provided.

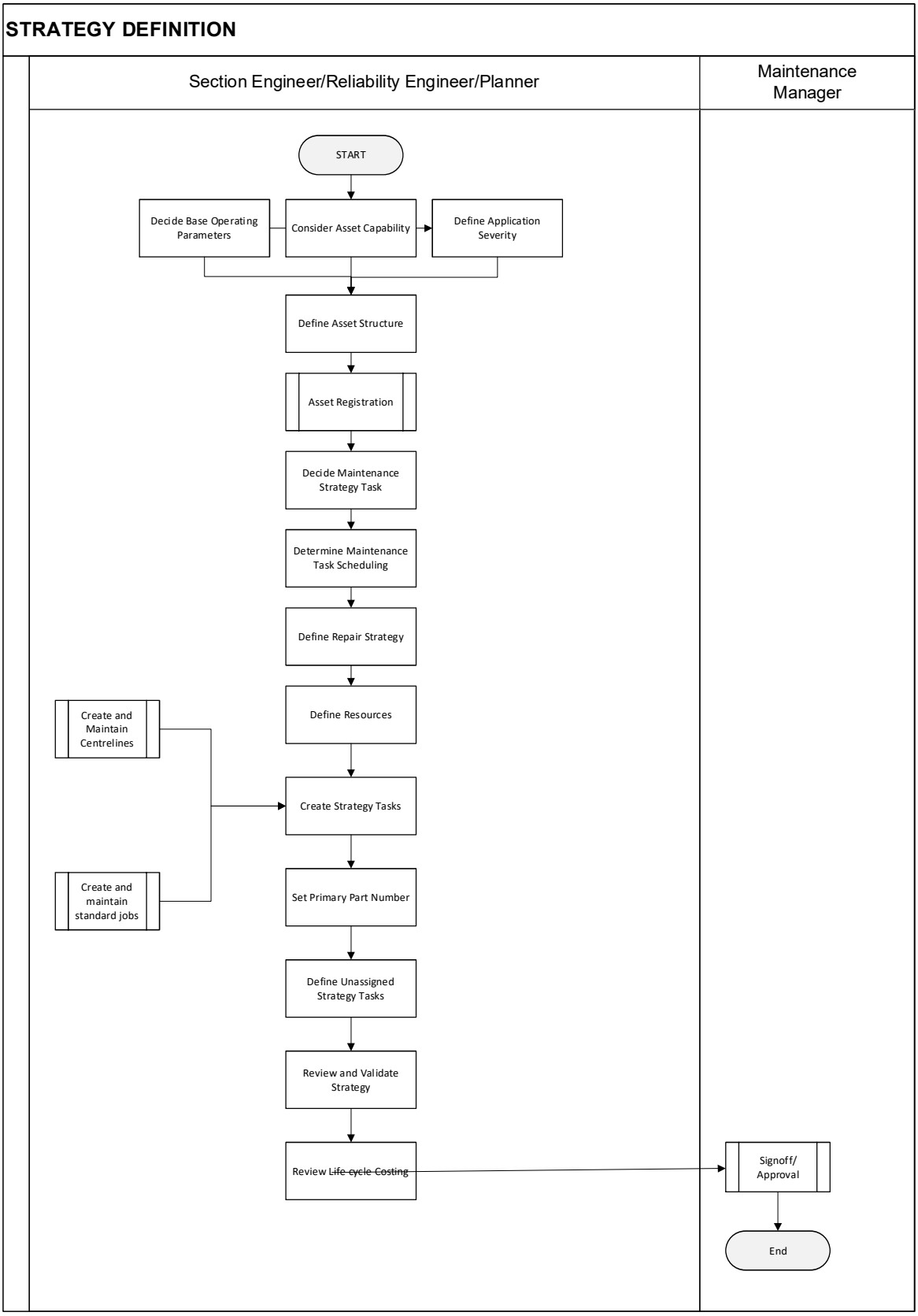
2.4. Strategy Definition

Strategy Definition is the process of defining strategy task against an AMT asset. Strategy Tasks are the basis for the asset's life cycle costing (LCC) and drives forecasting.

The objectives of the process are:

Objective	Importance/Benefit
Build an asset's life-cycle cost in AMT.	Enables AMT to forecast the asset's LCC. Enables AMT to generate planned work orders.

Figure 5 Strategy Definition Process Flow Chart



Process steps performed using AMT:

Process Flow Step	AMT Function/Report	Notes
Copy Centreline	Add / Copy / Delete Projection	Requires an appropriate Centreline (or Modelling equipment) to exist
Create Strategy Tasks	Equipment Editor > Tasks	
Set Primary Part Number	Equipment Editor > Tasks > Strategy Task > Component	Optional
Define Unassigned Tasks	Strategy Task Manager	Edit as required (First, Frequency, Cost)
Review & Validate Strategy	Projection Details Projection Summary	
Review Life Cycle Costing	Equipment Editor > Tasks Cost Per UOM Cost Per UOM per Interval	
Sign off / Approval		Approve the reviewed LCC of an Asset.

The following outputs are expected from this process:

Output	Description
Full life-cycle cost projection for the asset	Generate a life cycle Project Summary and Projection Details reports
LCC which includes failure provisions	More accurate LCC for an asset in AMT
Direct maintenance labour forecast	Ability to generate from AMT a Labour forecast report
Availability forecast	Ability to generate from AMT an Availability forecast report
Component change out forecast	Ability to generate from AMT a Component forecast report

Process Triggers:

Creation and/or update of a strategy for a new or existing centreline or equipment which is relevant to AMT. The process will be manually maintained in AMT.

Key Solution decisions regarding the **Strategy Definition** process:

- AMT is the master for an asset's maintenance strategy.
- Strategy Tasks are created and maintained in AMT
    - Can be sourced from AMT Centrelines
    - Can be supported by AMT Standard Jobs
- Suppression will be used for major and minor services per Date or Usage based services.
- All strategies will be associated to work groups, cost centres every site will have multiple work groups associated with a site.
- Labour activities will be exported from ERP system including their codes into system tables.

- STO will be associated with tasks that require shutdown (Fixed plants), and the flagging of CAPEX will be done where necessary.
- Expense elements will have a one-to-one mapping with product classes in the ERP, this will eliminate the issue of not having a product class indicator on the Misc tab of AMT. Procurement of inventory items to be done on the parts tab and Services on the Misc tab.
- Non planning task can be created and their Frequency multiplier will be 33%
- All cost associated with strategy will be in US Dollars (USD)
- Access to change the strategy tasks will be granted to certain users by the Mopani system administrators.

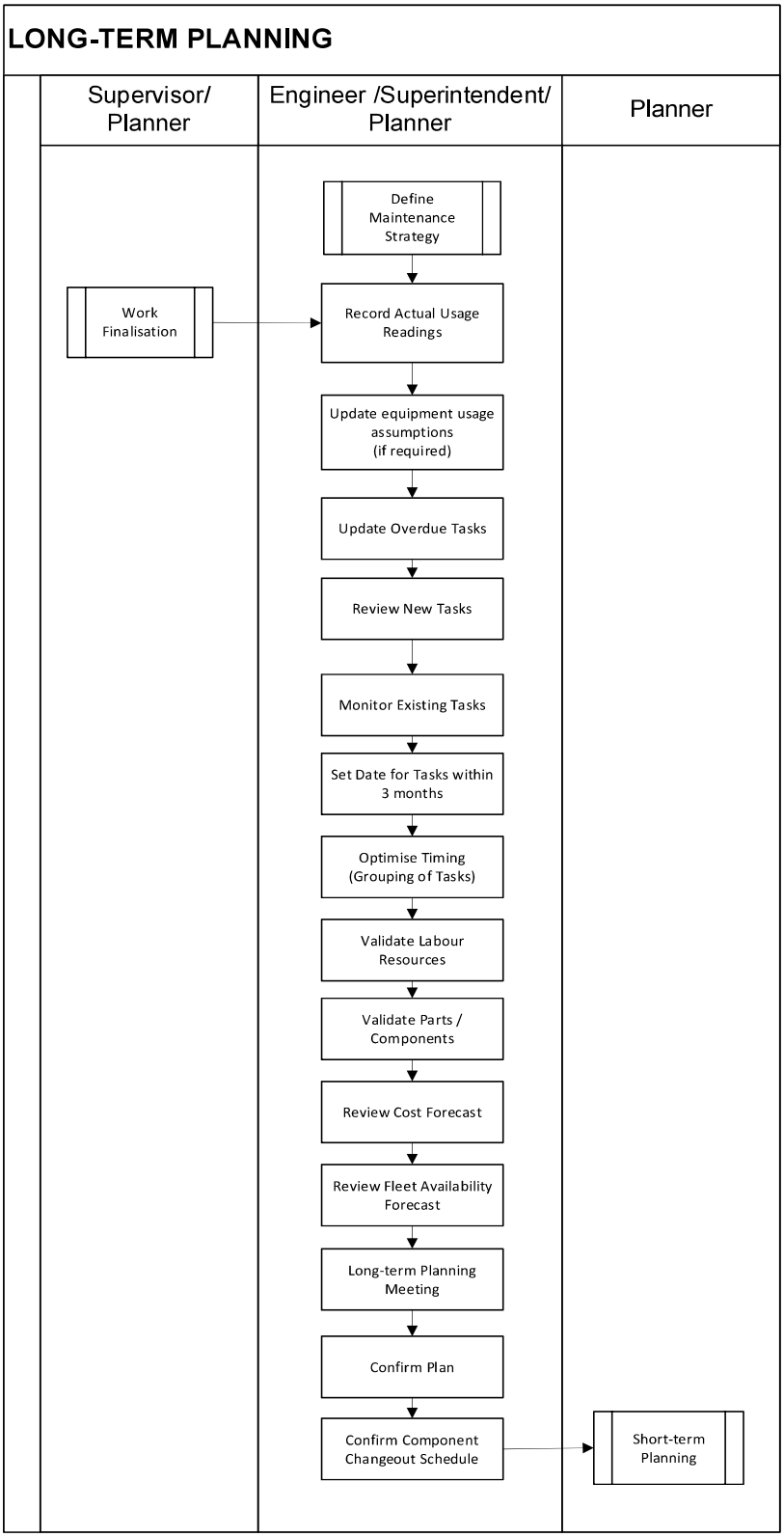
2.5. Long-term Planning

AMT's Long-term Planning functionality is designed to manage the Next Due of AMT strategy tasks in the long-term, i.e. 3 to 24 months into the future. This is usually before AMT has created a work order for the strategy task.

The objectives of the process are:

Objective	Importance/Benefit
Define a 2-year long-term plan that: • Optimises component lives • Optimises resources (Labour, components) • Meet availability targets	Improves the asset's LCC accuracy, which improves budgeting accuracy. Maximises component life and minimises component failure risk. Optimises scheduling efficiency. Reduce working capital. Coordinates maintenance activities with supporting departments & vendors

Figure 6 Long-term Planning Process Flow Chart



Process steps performed using AMT:

Process Flow Step	AMT Function/Report	Notes
Review Actual Usage Readings	Actual Usage Readings Quick Watch > No Recent Usage Reading	Future integration with external system.
Update equipment usage assumptions	Equipment Editor > Term Equipment > Editor Term > Usage Profile	For Simple utilisation For Advanced utilisation
Update Overdue Tasks	Quick Watch > Overdue Strategy Tasks Long-term Planning > Overdue = Yes	After updating Next Due, change Reviewed Status = Reviewed
Review New Tasks	Long-term Planning > Reviewed = No	Change status to Reviewed
Monitor Existing Tasks	Long-term Planning > Reviewed = Yes	
Review Tasks due within next 3 months	Long-term Planning > To (date)	After updating Next Due, Change Reviewed Status = Reviewed
Optimise Timing (Grouping of Tasks)	Long-term Planning > Gantt Chart	
Review Labour Forecast	Labour Forecast - Hours	
Review Parts / Component Forecast	Change out Analysis – Detailed Component status Parts Forecast Manager Parts Forecast	
Review Cost Forecast	Forecast Cost Worksheet Budget Forecast Performance Life Cycle Cost Summary	
Review Fleet Availability Forecast	Forecast Availability	
Confirm Component Schedule	Long-term Planning	Change Reviewed Status = Confirmed

The following outputs are expected from this process:

Output	Description
Rescheduled strategy tasks	Overdue tasks reforecast with a more accurate Next Due target
Optimise Next Due timings	Based on asset availability, resource availability, etc.
More accurate forecasts	Component change outs, parts, costs, labour, availability

Process Triggers

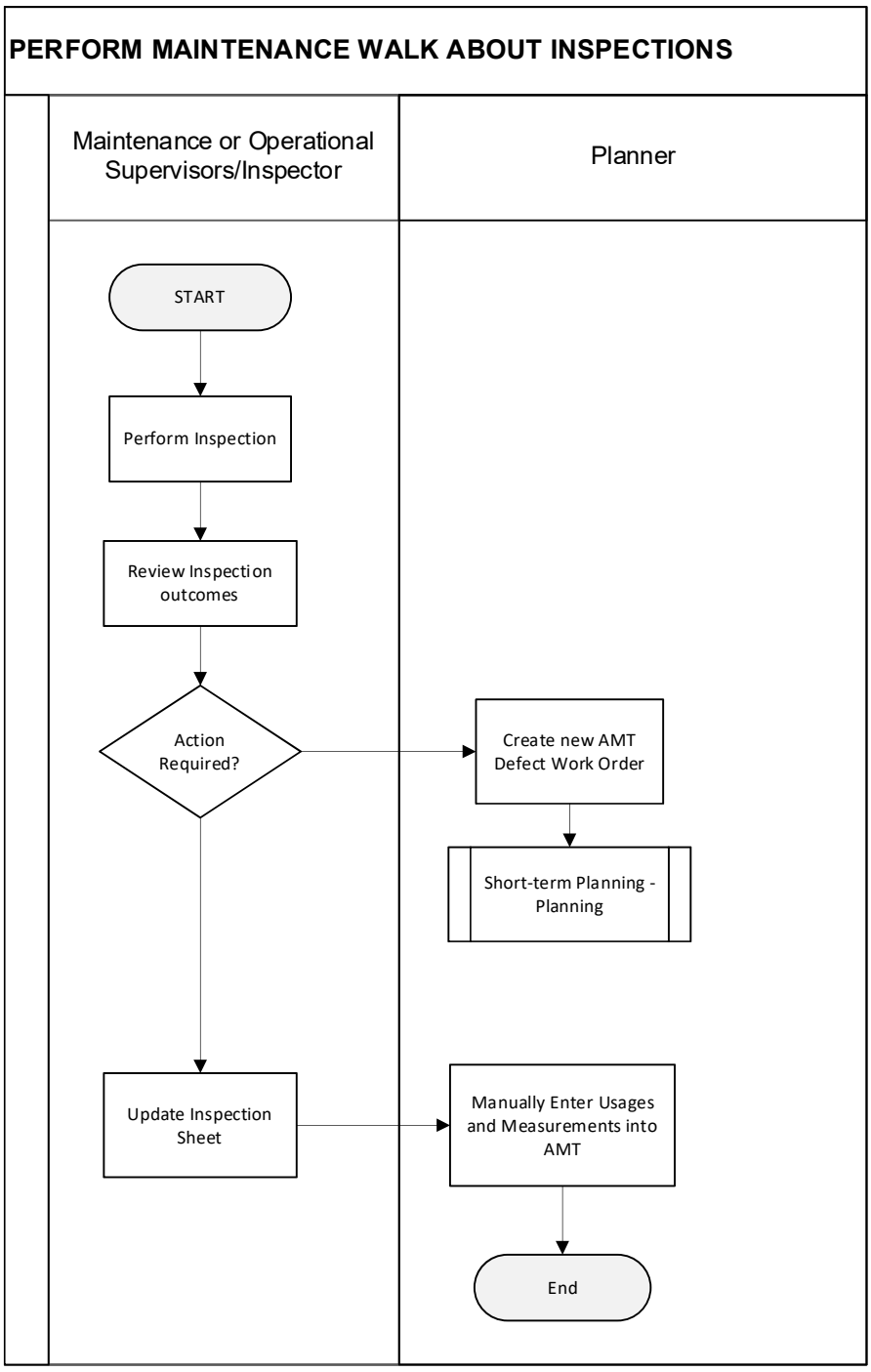
Need to plan work as a planner, long term planner or reliability engineer.

Key Solution decisions regarding the **Long-term Planning** process:

- Long-term Planning will be performed in AMT
- Actual usage readings will be recorded directly in AMT, will also be updated through the classic interface by csv file drop and through integration to the External Fleet Management Systems.

2.6. Perform a Maintenance Inspection

Figure 7 Perform a Maintenance Inspection Process Flow Chart



Process steps performed using AMT:

Process Flow Step	AMT Function/Report	Notes
Create new AMT Defect Work Order /	Equipment Status> Planning>Add New Work order	Will be fed into the list of currently outstanding backlogs and planned/performed as other work orders

Process Flow Step	AMT Function/Report	Notes
Create AMT Work Order\		

Outputs from this process are:

Output	Description
Inspection issues	Items which fail an inspection are visible in the Inspection window (along with inspection items which passed). Follow up repairs identified from inspections are created as AMT work orders.
Notifications	Condition monitoring alerts, which need to be investigated and actioned by the reliability team.
Work Orders	Approved maintenance interventions that are to be actioned by the planner.

Key Solution decisions regarding the **Perform a Maintenance Inspection** process:

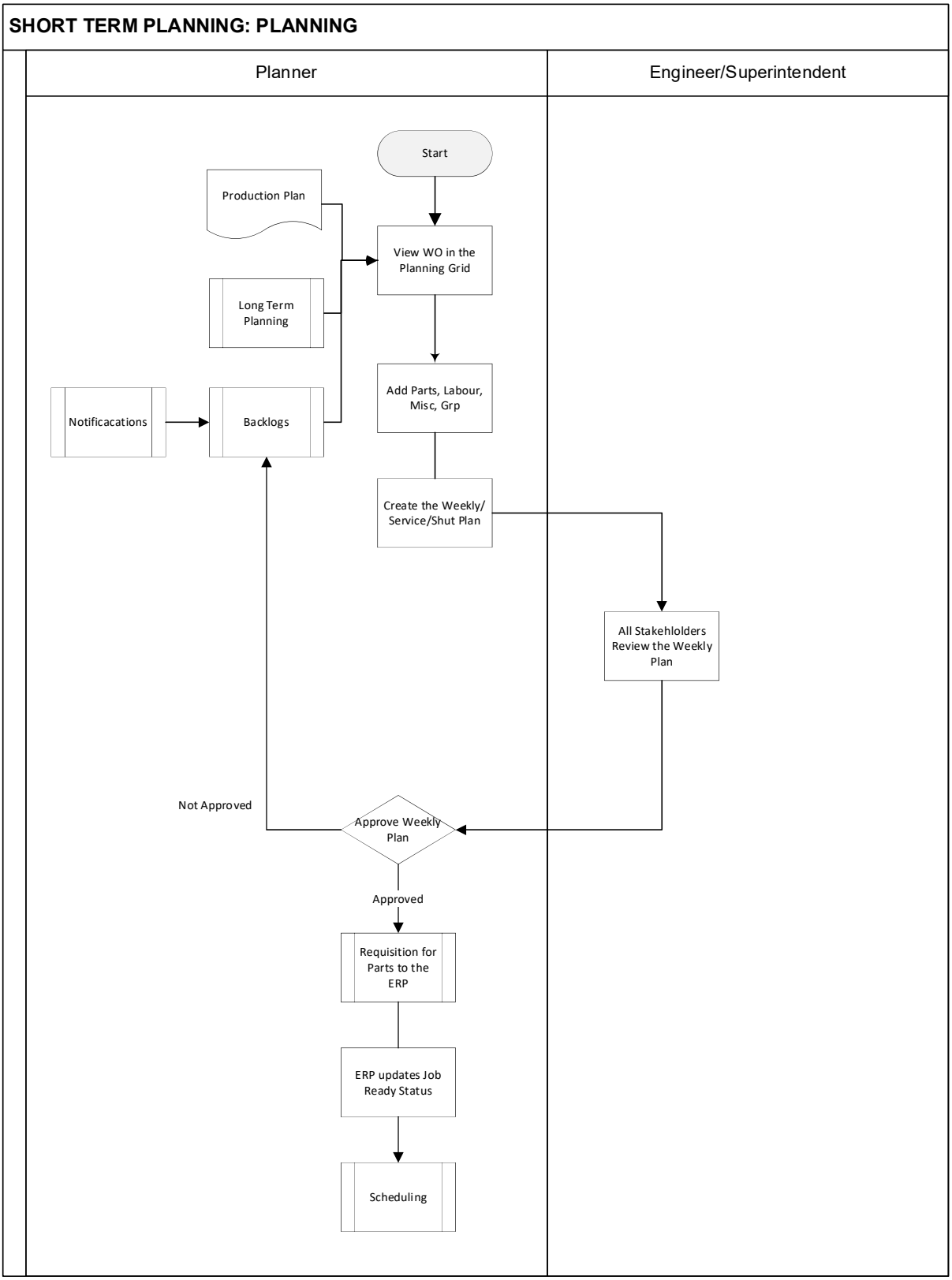
- Existing manual inspection forms/checklists will continue to be used by MOPANI until AMT Mobile is introduced in Phase 2. Daily, weekly, monthly and other inspections will be loaded as strategy tasks and checklists will be attached to relevant strategy tasks.
- Actual condition monitoring measurements will be entered manually into AMT Until external systems are integrated with AMT.
- Only approved defects raised from daily, weekly, and monthly inspections will be recorded into AMT.

2.7. Short-term Planning: Planning

The objectives of the process are:

Objective	Importance/Benefit
To define and organise the resources required to perform work orders.	Enables the planner to schedule work knowing the materials, labour and resources required to perform the work are available.
Confirm priorities	Ensure work is prioritised appropriately
Execute the Long-term Plan	Action the decisions made in the Long-term Planning process

Figure 8 Short-term Planning - Planning Process Flow Chart



Process steps performed using AMT:

Process Flow Step	AMT Function/Report	Notes
View Workorders Long Term Planning	Long Term Planning > Long Term Planning	Review confirmed component change out schedule from long term planning.
View Workorders in STP Planning grid.	Daily Planning > Planning	Filter for "Outstanding" work order
Add parts to the work order	Daily Planning > Edit Work Order > Parts	Additional parts as needed, mostly for work orders not based on Standard Jobs
Create Weekly Plan	Run Daily/Weekly Plan Report	
Update job ready status	Planning > Edit Work Order > Base Details	

Outputs from this process are:

Output	Description
Open and authorised work orders	AMT work orders created with planned parts, labour and misc. resource requirements defined.
Purchase Orders / Requisitions Created in ERP	Purchase Orders / Requisitions for parts and/or other miscellaneous resources (e.g. crane hire, subcontract labour) required for the work order are generated in AMT.

Key Solution decisions regarding the **Short-term Planning - Planning** process:

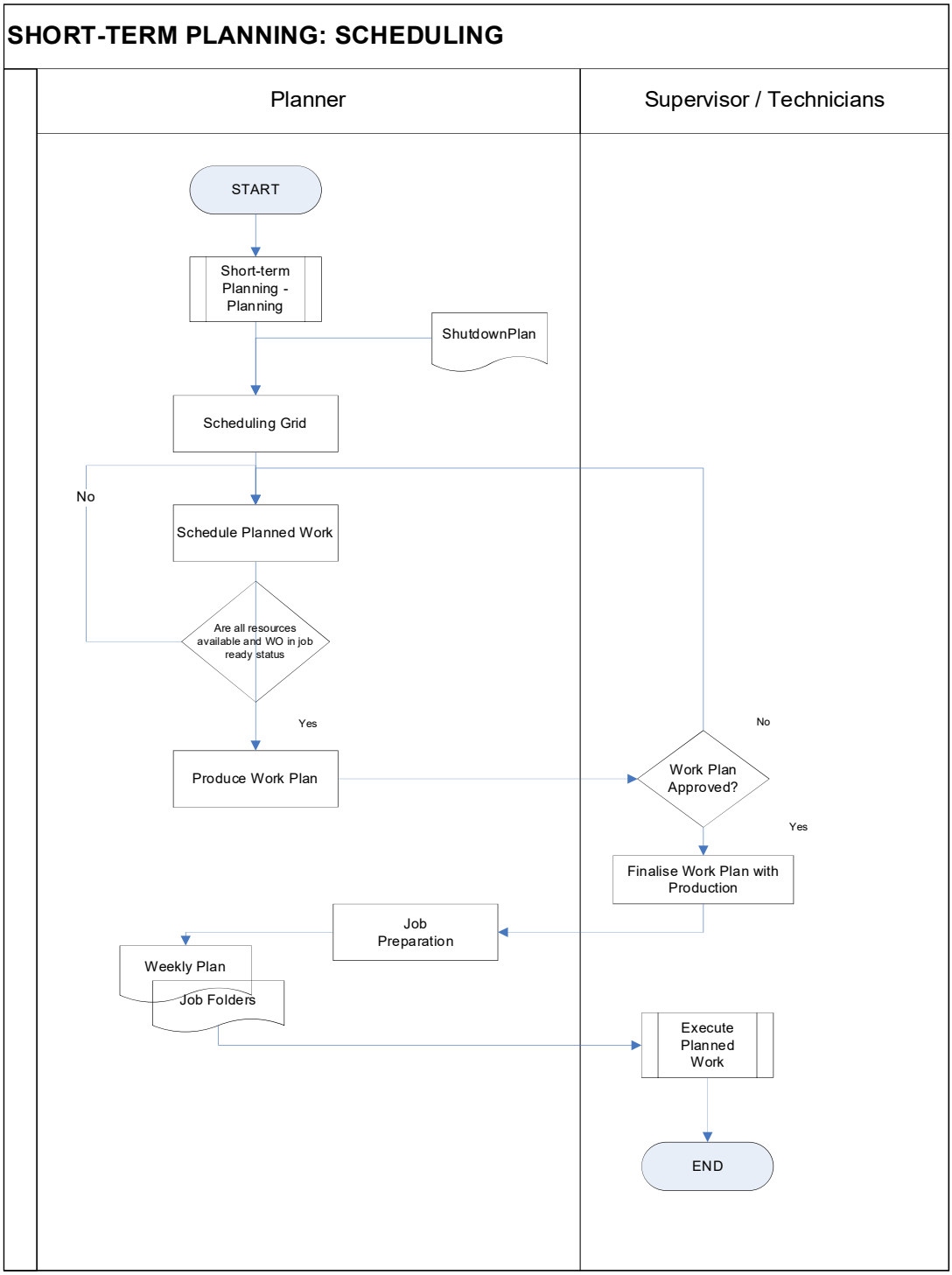
- Short-term Planning – Planning will be performed in AMT
- All ad-hoc maintenance work is recorded in AMT.
- Planners will control the planning details of component and non-component change out work orders.
- Work order will be auto authorised in AMT and will follow the Orbfusion authorisation process.
- Planned Parts, Labour and Miscellaneous items will be captured on the AMT workorder.
- Job ready status and parts status will be updated via integration.
- Labour capacity planning will be at the workgroup level.
- Work order will be printed and will form part of the documentation required in the requisition process.
- Purchase Orders / Requisitions for parts and/or other miscellaneous resources (e.g. crane hire, subcontract labour) required to execute an AMT work order will be created in ERP through integration.
- The workorder number will be generated from AMT with PR reference attachment and passed through to the ERP system via integration.

2.8. Short-term Planning: Scheduling

The objectives of the process are:

Objective	Importance/Benefit
Schedule work efficiently	Define an efficient work plan based on material and resource availability
Optimise downtime	Define a work plan which optimises downtime
To create and maintain weekly plans	Enables planners to communicate to all the stakeholders the work that needs to be performed, the following week.

Figure 9 Short-term Planning - Scheduling Process Flow Chart



Process steps performed using AMT:

Process Flow Step	AMT Function/Report	Notes
Schedule Planned Work	Daily Planning > Planning Daily Planning > Scheduling	Events will only be used for equipment which will report availability and reliability KPIs.
Resource Assignment and Availability Validation	Set Available Man Hours Daily Planning > Scheduling > Work Groups Resource Availability & Fill – Work Group	It is assumed Work Group labour validation is performed.
Product a Work Plan	Daily/Weekly Plan WorkPlan 2 WorkPlan – Export to MS Project	**NOTE** changes made in MS Project are not imported back into AMT. Any changes need to be made manually in AMT.
Job Preparation	Job Folders – Bulk Printing	

Outputs from this process are:

Output	Description
Daily/weekly work plans	Schedule of planned events and planned work orders per equipment
Shutdowns	Schedule for shutdowns produced
Job Folders	Job folder prepared for foreman to assign resources to execute work.

Key Solution decisions regarding the **Short-term Planning - Scheduling** process:

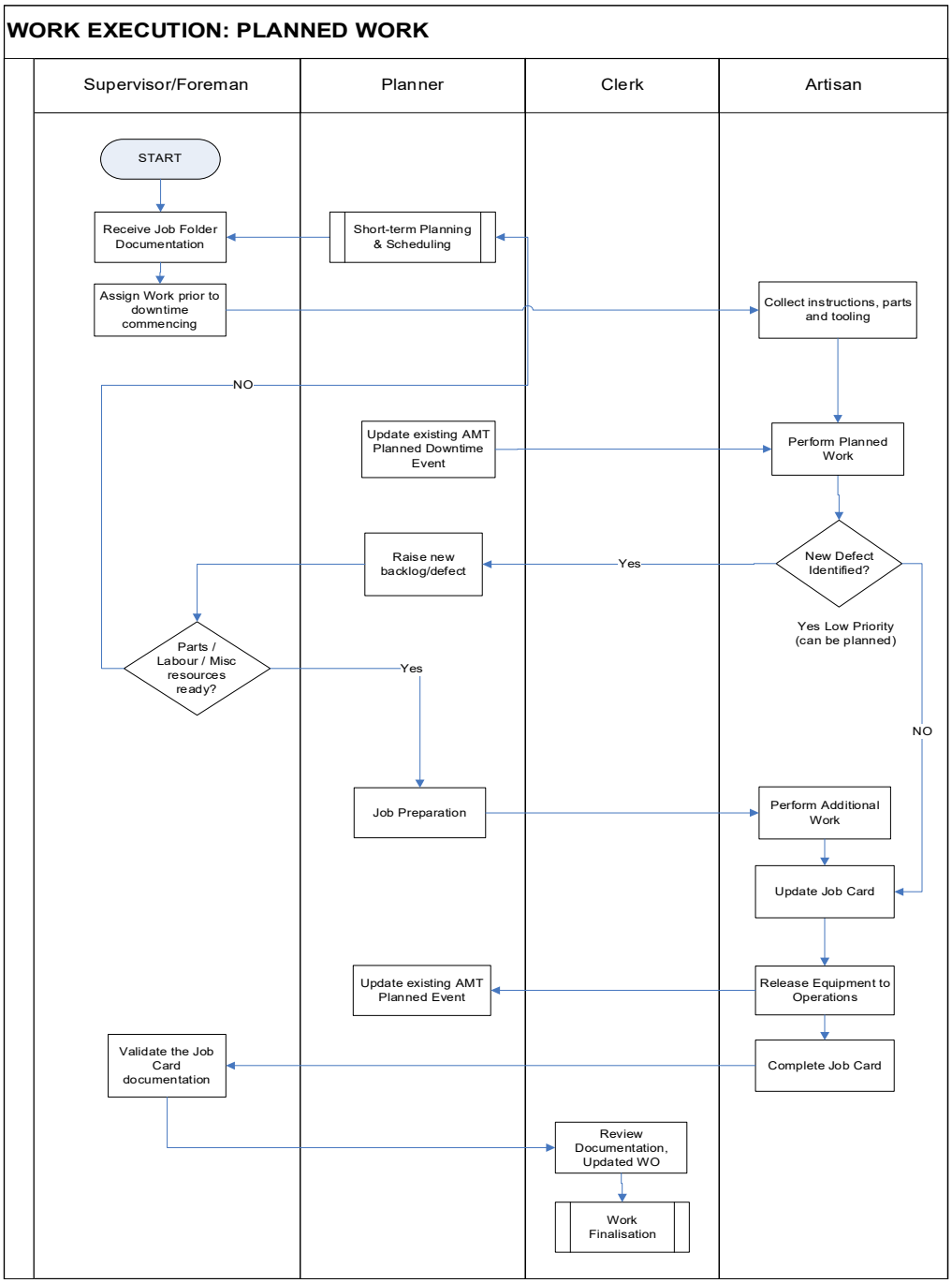
- Short-term Planning – Scheduling will be performed in AMT
- Work orders will be scheduled in AMT
- Labour resourcing will be performed in AMT at the Work Group level.
    - Users will maintain manually in AMT the available hours of each Work Group
- Work plans will be generated from AMT and exported to MS Project if required
    - Any changes to MS Project Plans will be manually updated in AMT
- Job Folders
    - Event scopes and Work scopes will be printed from AMT
    - Purchase Orders / Requisitions will be printed from ERP

2.9. Work Execution: Planned Work

The objectives of the process are:

Objective	Importance/Benefit
To perform planned work	Update AMT with actuals for work performed

Figure 10 Work Execution – Planned Work Process Flow Chart



Process steps performed using AMT:

Process Flow Step	AMT Function/Report	Notes
Update existing AMT Planned Downtime Event	Advanced Planning equipment • Equipment Status > Shift log > Update Event • Equipment Status > Shift log > Work Order Standard Planning equipment: • Equipment Status > Workorder	
Raise New Defects/Backlog in AMT	Equipment Status > Workorders	
Job Preparation	Equipment Status > Shift log > Add Blank Work Order to Event	Add the defect work order to the planned event, then: 1. Update work order's codes (component code, modifier code, task type code) 2. Update work order's operation financial codes (cost centre, expense element) 3. Add parts / labour / misc. items to the work order as required 4. Create Purchase Order / Requisition in ERP (if required)
Update AMT Planned Event	Advanced Planning equipment • Equipment Status > Shift log • Equipment Status > Shift log > Work Order Standard Planning equipment: • Equipment Status > Workorder	

Outputs from this process are:

Output	Description
Actual downtime for planned work	Actual downtime for planned work is entered into AMT, which enables Planned vs. Actual Downtime to be compared, %Planned downtime to be reported, %Planned downtime events (occurrences) to be reported, etc.
Planned work order(s) updated	Planned work orders, within the planned event, will be updated (completed, deferred, or abandoned). Those planned work orders which are set to Completed status, and which are linked to strategy tasks, will result in the strategy tasks being rescheduled (note, rescheduling of a strategy task will also occur if the work order's status is In Progress).
Additional Defect work order(s)	Additional defect work orders that were added and executed during the planned downtime event will form part of the equipment's total maintenance history.

Key Solution decisions regarding the **Work Execution – Planned Work** process:

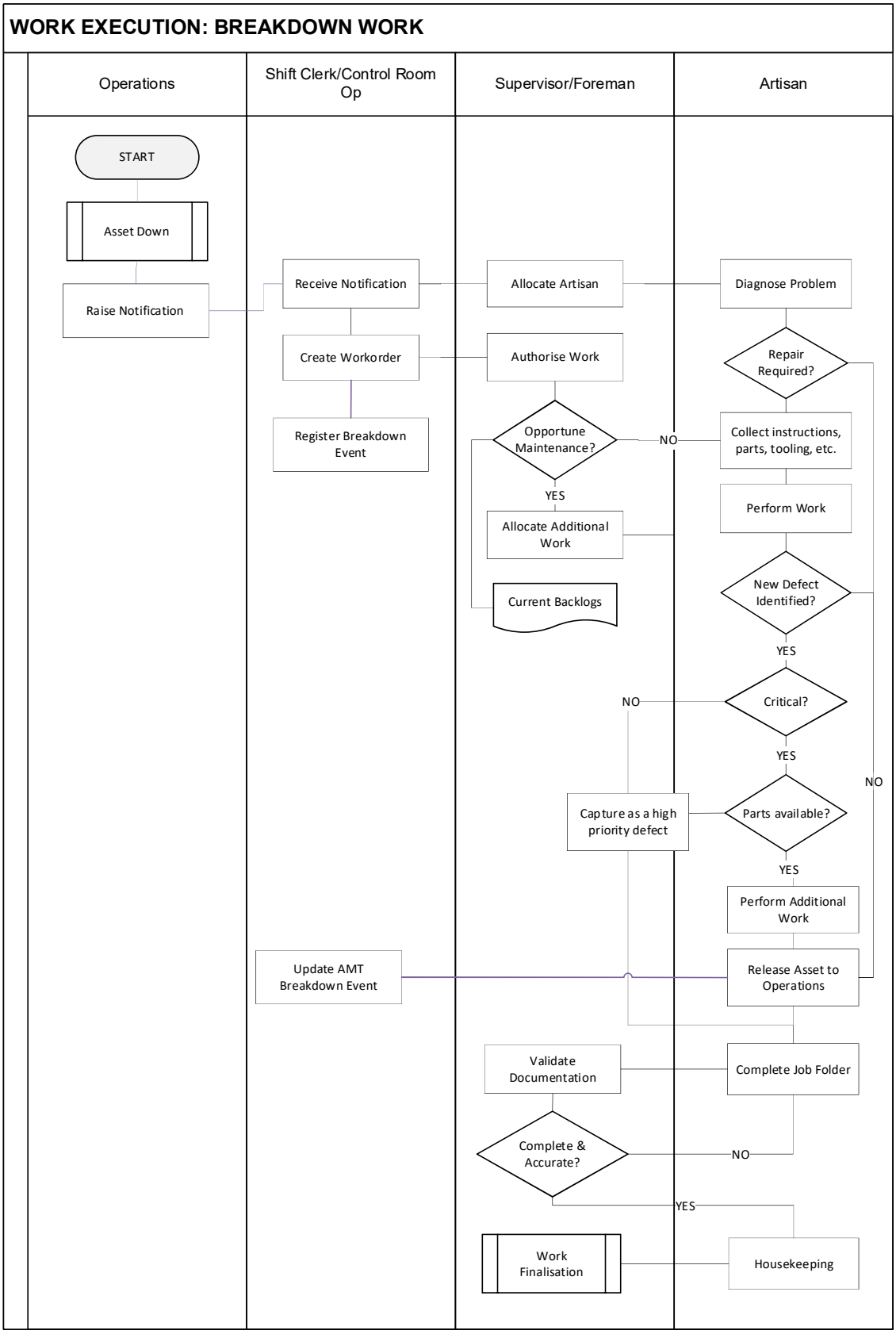
- Actual downtime will be recorded manually in AMT against existing Planned Events
- AMT will be used to capture downtime delays and downtime responsibility within AMT Events

2.10. Work Execution: Breakdown Work

The objectives of the process are:

Objective	Importance/Benefit
To perform breakdown work	Update AMT with actuals for work performed

Figure 11 Work Execution – Breakdown Work Process Flow Chart



Process steps performed using AMT:

Process Flow Step	AMT Function/Report	Notes
Create an AMT Breakdown Event	Advanced Planning equipment • Equipment Status > Shift log • Equipment Status > Shift log > Work Order Standard Planning equipment: • Equipment Status > Workorder	Only required if the equipment is configured in Advanced Planning mode. Steps to create a new work order: 1. Set work order's codes (component code, modifier code, task type code) 2. Set work order's operation financial codes (cost centre, expense element) 3. Add parts to the work order Note, if standard planning is used then place a check mark in the Breakdown box on the Feedback tab of the work order.
Job Preparation	Equipment Status > Shiftlog > Add Blank Work Order to Event	Add the defect work order to the breakdown event, then: 1. Update work order's codes (component code, modifier code, task type code) 2. Update work order's operation financial codes (cost centre, expense element) 3. Add parts to the work order 4. Create Purchase Order / Requisition in ERP (if required)
Update AMT Breakdown Event	Advanced Planning equipment • Equipment Status > Shift log • Equipment Status > Shift log > Work Order Standard Planning equipment: • Equipment Status > Workorder	Review the completeness and accuracy of event and work order details. For Standard Planning, update the Actual Start/End Time/Duration on the work order's Planning tab.

Outputs from this process are:

Output	Description
Actual downtime for breakdown work	Actual downtime for breakdown work is captured in AMT, which enables %Breakdown downtime to be reported, %Breakdown downtime events (occurrences) to be reported, Top 10 downtime causes to be analysed, etc.
Work order(s) updated	Work orders will be updated (completed, deferred, or abandoned). Those work orders which are set to Completed status, and which are linked to strategy tasks, will result in the strategy tasks being rescheduled (note, rescheduling of a strategy task will also occur if the work order's status is In Progress).
Additional Defect work order(s)	Additional defect work orders that were added and executed during the breakdown event, will form part of the equipment's total maintenance history.

Key Solution decisions regarding the **Work Execution – Breakdown Work** process:

- A breakdown Notification will be created during night shift by operations.
- Artisans will not be allowed to collect parts without work order number.
- The breakdown parts requisitioning will follow the Mopani Business Process.

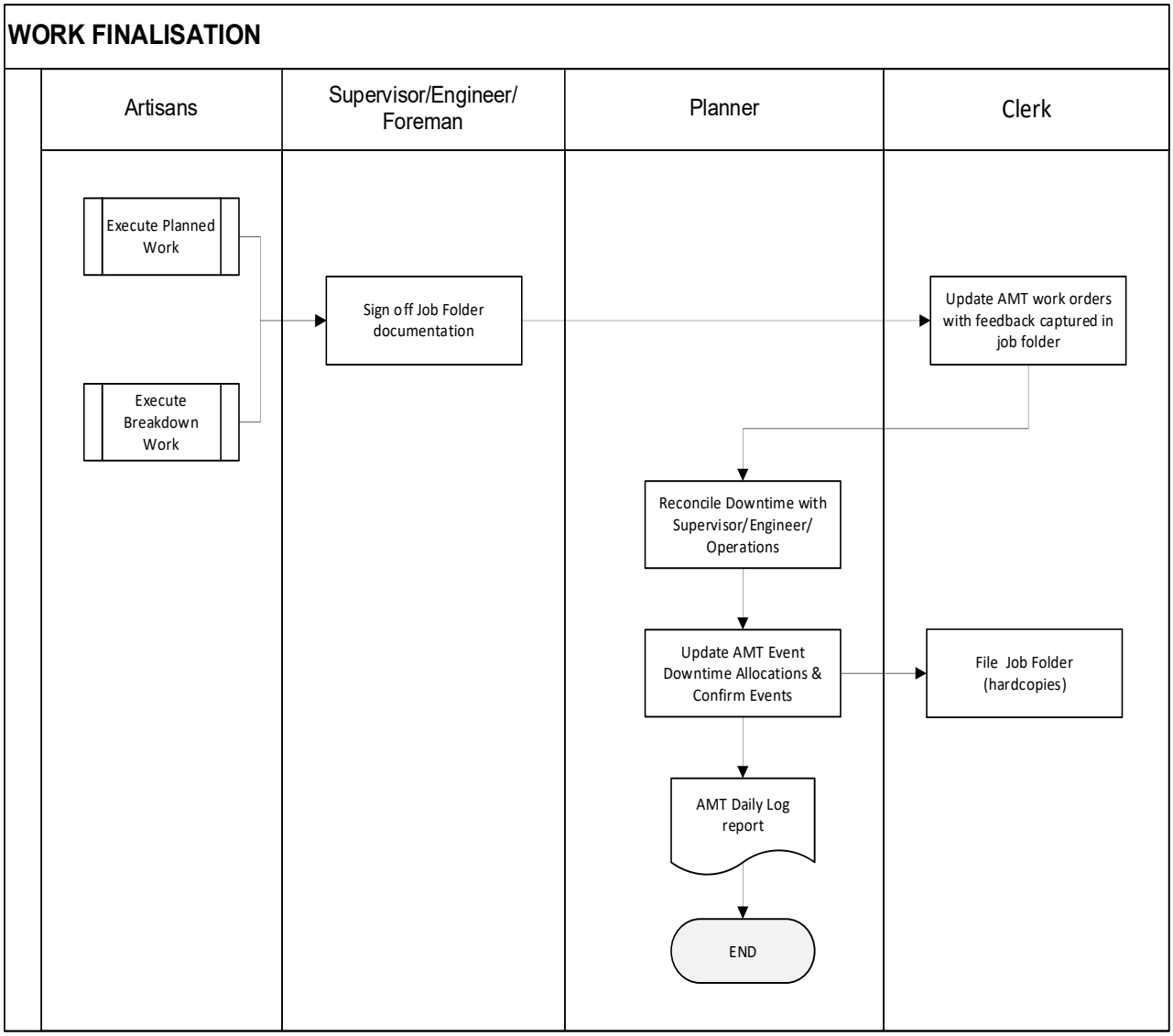
- Actual downtime and uptime recorded in AMT for breakdown maintenance downtime.
- AMT will be used to capture downtime delays and downtime responsibility within AMT Event.
- Mindola Shift clerk will raise the breakdown work order.
- Other sites: Decision to be made on who will create the breakdown work order.
- Planner will be responsible to ensure the Event and Work order captured overnight are correct.
- Notifications will be created as part of the Breakdown Process.

2.11. Work Finalisation

The objectives of the process are:

Objective	Importance/Benefit
Capture additional details for work executed	Enables AMT to perform analysis on feedback provided on work executed
Reconcile downtime	Reconcile maintenance's downtime log with production's downtime log and agree responsibility for actual downtime incurred.

Figure 12 Work Finalisation Process Flow Chart



Process steps performed using AMT:

Process Flow Step	AMT Function/Report	Notes
Update AMT Workorders with feedback captured in job folder.	Equipment Status > Events Edit Work Order	
Reconcile Downtime with Dispatch/Operations	Equipment Status > Events > Edit Event Daily Downtime Log Shift Downtime Log	Changes are to be made and updated AMT.
Update AMT Event Downtime Allocations and Confirm Events	Equipment Status > Events	

Outputs from this process are:

Output	Description
Downtime reconciled with Operations	Maintenance downtime in AMT reconciles with Operations maintenance downtime values.
Downtime Delays captured	AMT Events' Downtime Allocation(s) populated with downtime activities, delays, causes of delays, downtime responsibility. All of which is critical for accurate downtime related KPIs and analysis. AMT Event is "confirmed."
Equipment performance	Various Equipment KPI reporting will be available for users.

Key Solution decisions regarding the **Work Finalisation** process:

- Once the planned/breakdown work is complete, all actual maintenance details will be captured in AMT (e.g. symptom/cause of failure, repair details)
    - Work order maintenance completion occurs in AMT
    - Work order financial closure occurs in ERP
    - ERP will provide AMT with the actual cost of the planned/breakdown maintenance work performed per AMT work order where parts or miscellaneous items was used
        - Source data is ERP Purchase Orders and Requisitions for Parts
        - Includes the actual cost bearer(s), cost centre(s) and expense elements to which the actual costs have been transacted
        - The work order settlements will be imported into AMT via the classic interface and will be the source of cost reports
- Actual productive labour hours required to execute an AMT work order will be manually recorded on the AMT work order.
- If actual downtime (i.e. down date: time / update: time) is to be changed based on downtime reconciliation discussions with Operations, then the change will be made manually in AMT.

2.12. Shutdown Planning

The objectives of the process are:

Objective	Importance/Benefit
To show the Shutdown process	Understanding of shutdown process within existing processes
Shutdown Planning	Enable the planner to schedule work knowing the materials, labour and resources required to perform the work are available for the Shutdown

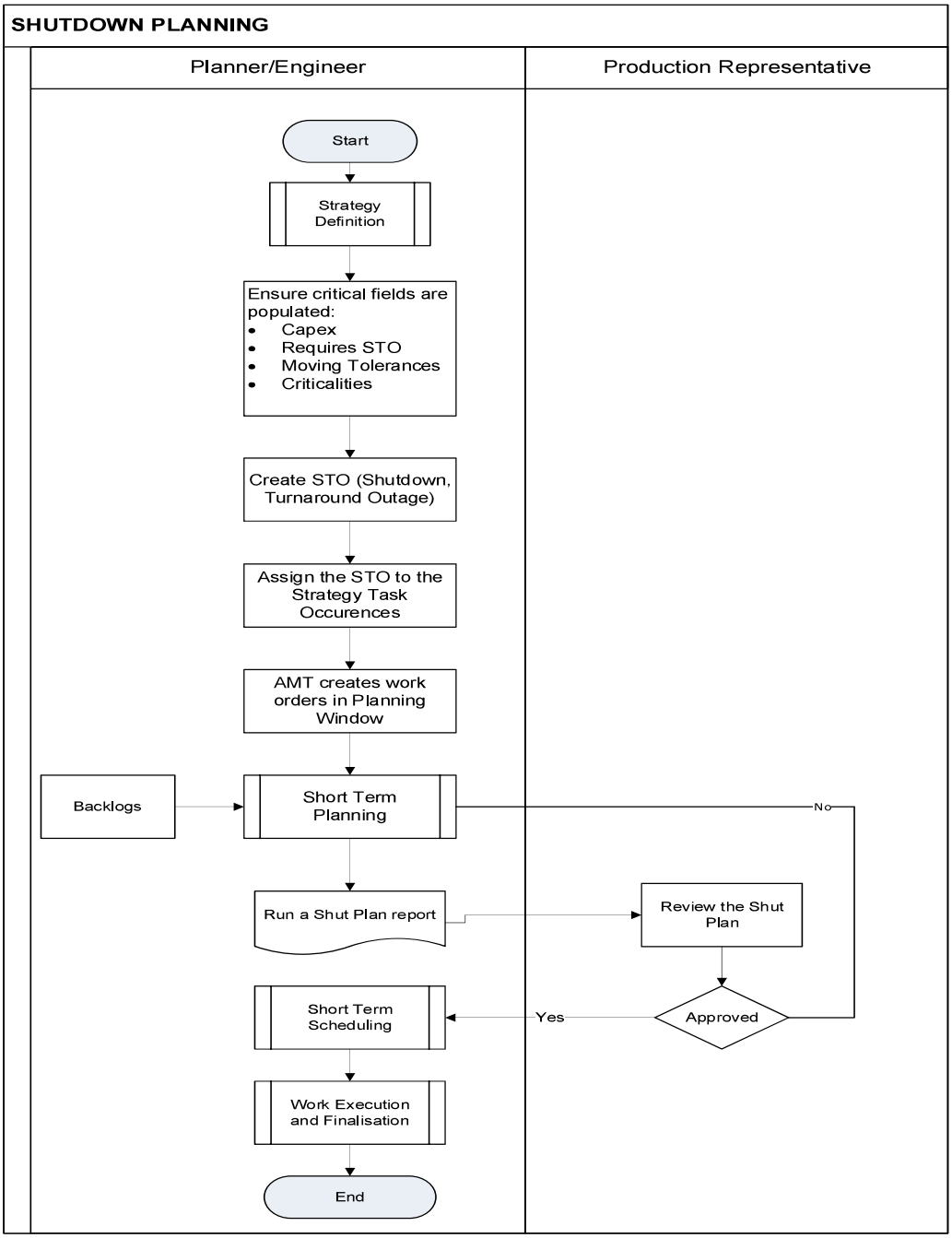
Key Solution decisions regarding the **Shut Down Planning process:**

- A unique job number for the Shutdown will be created in the ERP.
- The job number will be created as an STO in AMT. (Job Number may be the STO External ID)

Outputs from this process are:

Output	Description
Shutdown Planning	Scheduled occurrences assigned to STO.
Shutdown Budget	Creation of the Budget for Shutdowns

Figure 13 Shutdown Planning Process Flow Chart



Process steps performed using AMT:

Process Flow Step	AMT Function/Report	Notes
Strategy Definition	Equipment Administration >Tasks	Strategy Tasks for Equipment
Ensure critical fields are populated: • Capex	Strategy Task Form	Ensure all critical fields are populated

Process Flow Step	AMT Function/Report	Notes
• Requires STO • Moving Tolerances Criticalities		
Create STO (Shutdown, Turnaround, Outage)	STO Manager > STO Category STO Manager > Create STO	Define naming and when the STO will happen
Assign the STO to the Strategy Task Occurrences	Long Term Planning Advanced	
AMT creates work orders in Planning Window	System Task Manager	AMT will automatically create the work orders from the Strategy Tasks when they are due (within the planning window)
Short Term Planning	Planning > Planning	See Short Term Planning Process
Backlogs	Planning > Planning	Backlogs to be done during the Shutdown will be assigned the STO number in Short Term Planning
Run a Shut Plan Report	Run AMT Report	Run Daily Reports, e.g., Work Plan 2 report
Review the Shut Plan		Planner and Production Review the shut Plan and make necessary adjustments.
Approved		Shut Plan approved outside AMT
Short Term Scheduling	Planning > Scheduling	Work orders are scheduled by assigning events
Work Execution and Finalisation		See Work Execution and Finalisation Process

2.13. Strategy Optimisation

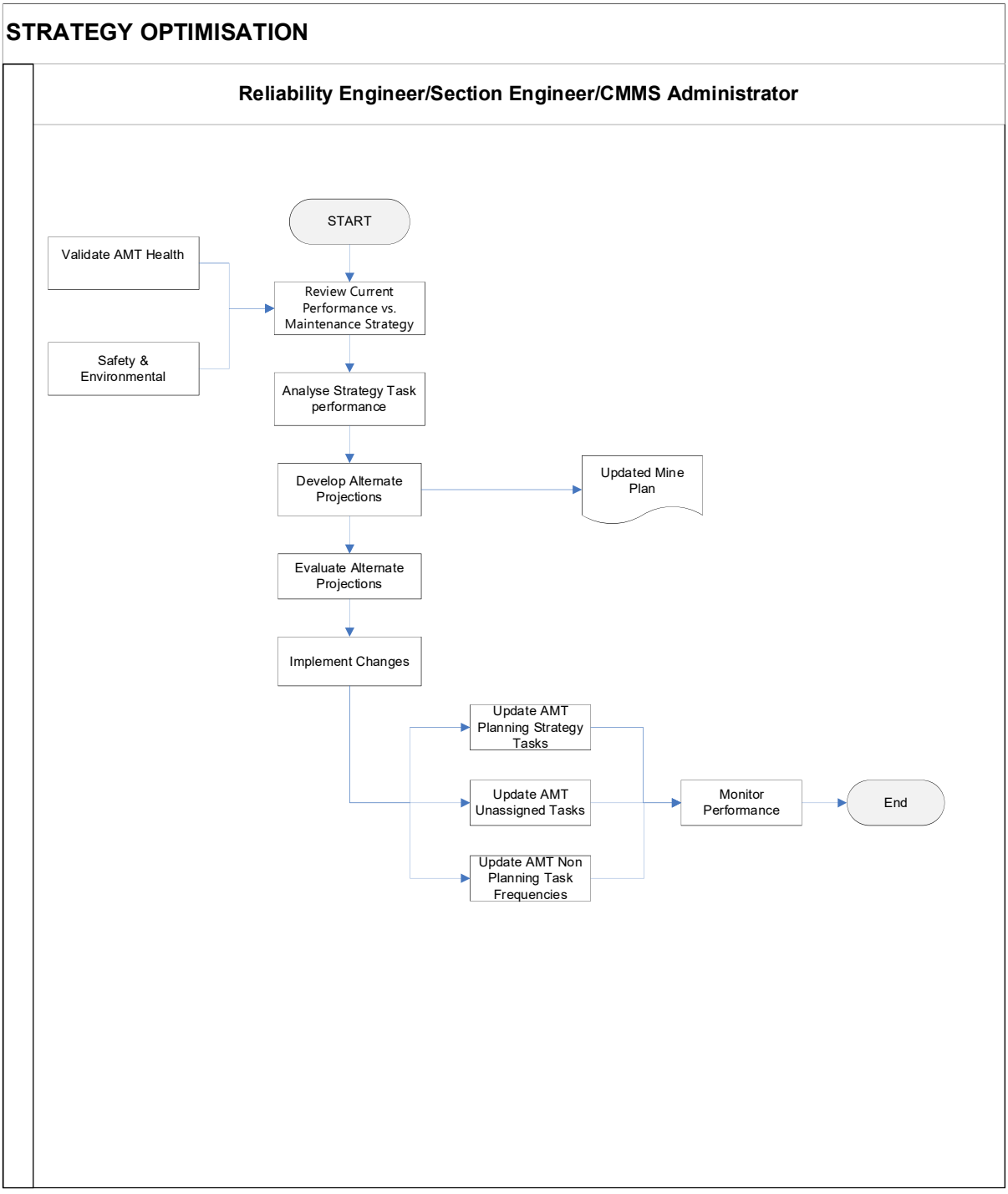
The objectives of the process are:

Objective	Importance/Benefit
To optimise asset performance and minimise operating costs	Increase availability. Increase reliability. Improve component performance. Reduce risk (cost, productivity, safety, failure). Reduce maintenance costs. Reduce working capital.
To direct continuous improvement projects	Focus limited resources on improvement projects, which will maximise return on investment (return is measured by improved availability, reliability, and lower life-cycle costs).
Expand maintenance strategy coverage	Increase coverage and refine quality of standard jobs and strategy tasks in AMT.

The objectives of the process are:

Objective	Importance/Benefit
To optimise asset performance and minimise operating costs	Increase availability. Increase reliability. Improve component performance. Reduce risk (cost, productivity, safety, failure). Reduce maintenance costs. Reduce working capital.
To direct continuous improvement projects	Focus limited resources on improvement projects, which will maximise return on investment (return is measured by improved availability, reliability, and lower life-cycle costs).
Expand maintenance strategy coverage	Increase coverage and refine quality of standard jobs and strategy tasks in AMT.

Figure 14 Strategy Optimisation Process Flow Chart



Process steps performed using AMT

Process Flow Step	AMT Functionality	Notes
Validate AMT Health	Quick Watch screen – No Recent Usages Overdue Strategy Tasks report	

Process Flow Step	AMT Functionality	Notes
	Work Order Settlement Manager screen	
Review current performance	Cost Per UOM – By Date report Cost Per UOM – By Usage report Cost Per UOM – Comparison report Cost Per UOM Per Interval report PCV Chart (if budget life cycle projections exist) report Cost Analysis – Life Cycle report Usage vs. Profile Snapshot report	Budget life cycle projections are not mandatory. These may only be populated for select Equipment by users if required.
Analyse Strategy Task Performance	PCV Analysis report	PCV (Projected Cost Variance) compares one projection to another, e.g., comparing a Current projection to an Alternate projection or to a Budget life cycle projection. Note, Budget life cycle projections are not mandatory. These may only be created and populated for select Equipment by users if required.
	Strategy Task Analyser screen Drill down using additional AMT functionality: • Life Analysis: Component Life Distribution report • Cost Analysis: Cost Analysis - Life Cycle report • Labour Analysis: Labour Hours Variance Analysis report	Note, Duration comparisons are not possible due to AMT's work management functionality not being deployed.
Develop Alternate Projections	Add / Copy / Delete Projection screen	
	Multi Scope Editor screen	This editor is required to make the following type of changes to strategy tasks linked to standard jobs: • Parts Cost % Factor • Miscellaneous % Factor • Duration % Factor Comments/Reasons for the change are saved.
	Multi-Task Editor screen	This editor is required to make the following changes: • First / Frequency • Parts Cost • Misc Cost • Duration • Work group • Labour Hours

Process Flow Step	AMT Functionality	Notes
Evaluate Alternate Projections	What If – Life Cycle Analysis report What If – Period Analysis report What If – Cost Per UOM report What If – Cost Analysis report RVIC Chart report High Risk report High Opportunity report Forecast Availability report. Labour Forecast – Hours report Cost Per Interval report Cost Per UOM Per Interval report	The What If reports listed allow a side-by-side comparison. Other reports listed either can be run for a nominated projection or can compare a side-by-side comparison; depending on the report's capability.
Update AMT Planning Strategy Tasks	Multi-Task Editor screen	Parameter(s): Frequency, Parts Cost, Misc Cost
Update AMT Unassigned Tasks	Multi-Task Editor screen	Parameter(s): Frequency, Parts Cost, Misc Cost
Update AMT Non-Planning Task Frequencies	Multi-Task Editor screen	Parameter(s): First and/or Frequency
Monitor Performance	Quick Watch screen	

The following outputs are expected from this process:

Output	Notes
Updated maintenance strategy in AMT (strategy tasks, unassigned tasks, and standard jobs).	AMT will be updated automatically.
Accurate usage, cost, and resource forecasts.	Reports generated from AMT.
Projected cost savings (ROCE/ROI from the maintenance strategy change).	AMT What If Analysis reports.

Key Solution decisions regarding the **Strategy Optimisation** process:

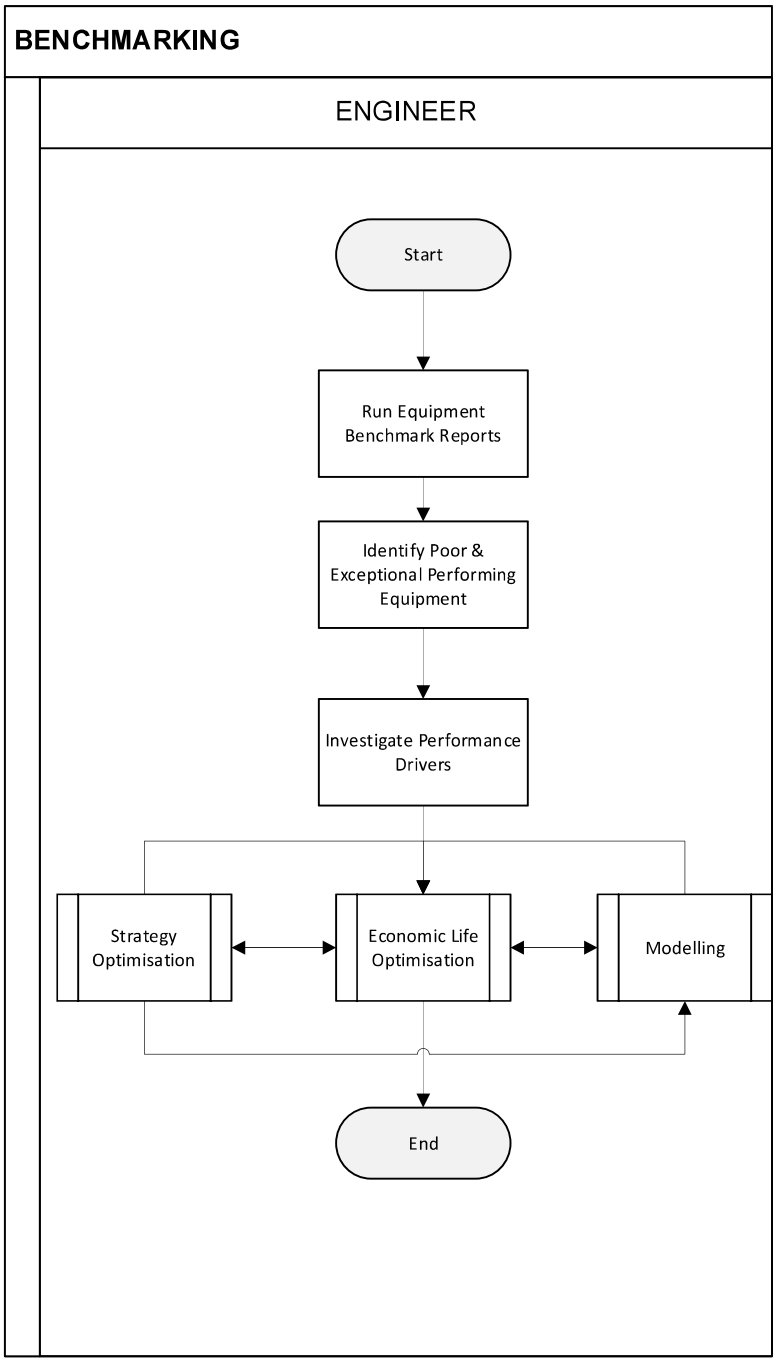
- Out-of-the-box AMT strategy optimisation functionality will be used, including Strategy Task Analyser, to identify strategy optimisation opportunities, and develop and compare "What If" alternate maintenance strategy options
- AMT will be used to create What If scenarios.

2.14. Benchmarking

The objectives of the process are:

Objective	Importance/Benefit
Compare Equipment performance within a single site and/or across multiple sites using Cost Per UOM metrics	Maintain Standards and optimise Equipment performance
Compare component performance within a single site and/or across multiple sites using Component Reliability Analysis	Maintain Standards and optimise Component performance
Compare consistency of maintenance strategies within a single site and/or across multiple sites to identify variances	Continuous improvement

Figure 15 Benchmarking.



Process steps performed using AMT

Process Flow Step	AMT Function/Report	Notes
Run Equipment Benchmark reports	Cost per Interval report Cost per Interval per UOM report Cost per Interval Comparison report Cost per UOM – By Date report Cost per UOM – By Usage report Cost per UOM – Comparison report Unassigned Cost per UOM – By Date Unassigned Cost per UOM – By Usage Unassigned Cost per UOM – Comparison Cost Analysis – Life Cycle report Unassigned Cost Analysis report Cost Trending	
Investigate Performance Drivers	Strategy Task Manager screen	Compare maintenance tactics consistency
	Strategy Task Analyser screen Component Life Distribution report	Compare performance vs. maintenance strategy assumptions
Strategy Optimisation	Refer to Strategy Optimisation Section	Refer to Strategy Optimisation Section
Economic Life Optimisation	Refer to Economic Life Optimisation Section	Refer to Economic Life Optimisation Section
Modelling	Refer to the Modelling Section	Refer to Modelling
	Cost per Interval report - Modelling Cost per UOM – By Date report - Modelling Cost per UOM – By Usage report - Modelling Cost per UOM – Comparison report - Modelling	Benchmarking reports used to compare different Modelling equipment (a Modelling equipment – which represents a future potential equipment – could replace the existing Real equipment in future).

The following outputs are expected from this process:

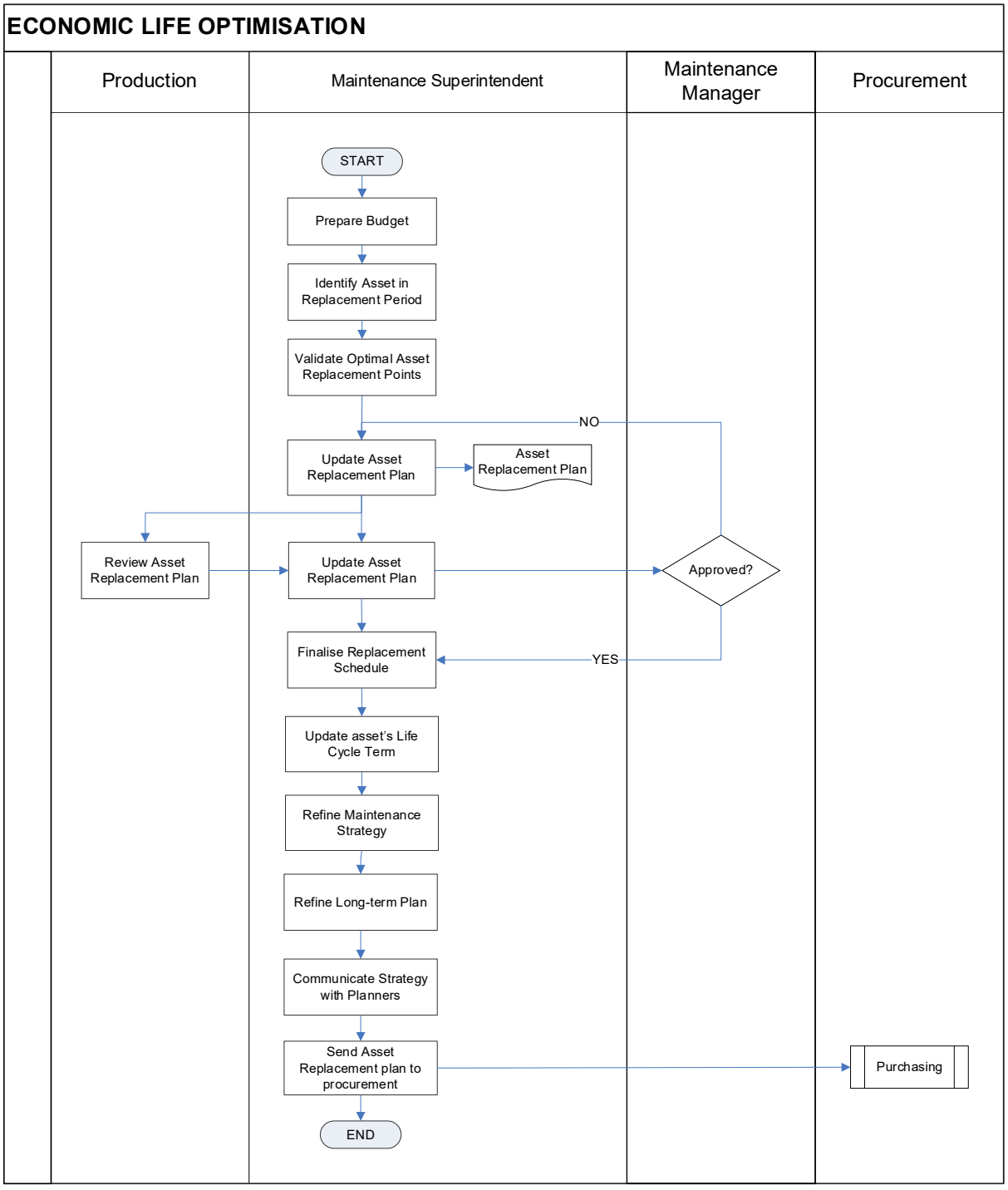
Output	Notes
Benchmark Performance Indicators	Identification of high performing vs. low performing equipment from a life cycle cost perspective, and underlying drivers of this performance which (a) Reliability and Planners may investigate more deeply, and/or (b) which Fleet Planners may use to determine which equipment should be retired.

2.15. Economic Life Optimisation

The objectives of the process are:

Objective	Importance/Benefit
To assist in the identification of Equipment to be disposed.	Operating costs are minimised
To identify optimal points to dispose of Equipment.	Operating costs are minimised
To ensure the maintenance strategy reflects Equipment replacement plan.	Operating costs are minimised
To correctly identify when a new Equipment is required (if the Equipment is to be replaced by new Equipment).	Accurate maintenance forecast

Figure 16 Economic Life Optimisation Process Flow Chart



Process steps performed using AMT

Process Flow Step	AMT Function/Report	Notes
Identify asset in replacement period	Cost per UOM reports Cost per Interval reports	Used to identify potential asset(s) to be replaced based on a benchmark comparison.
Validate Optimal Asset Replacement Points "when"	RVIC Chart RVIC analysis	Used to identify the optimal replacement point for the asset which will be disposed.
Update asset's life cycle term	Equipment Editor > Term	
Refine Maintenance Strategy		Refer to Strategy Optimisation process
Update asset's long-term plan		Refer to Long-term Planning process

The following outputs are expected from this process:

Output	Notes
Updated asset replacement plan	
Documented maintenance strategy of the asset which will be disposed	
Updated maintenance strategy of the asset which will be disposed	
Updated long-term plan of the asset which will be disposed	

2.16. Create Modelling Assets

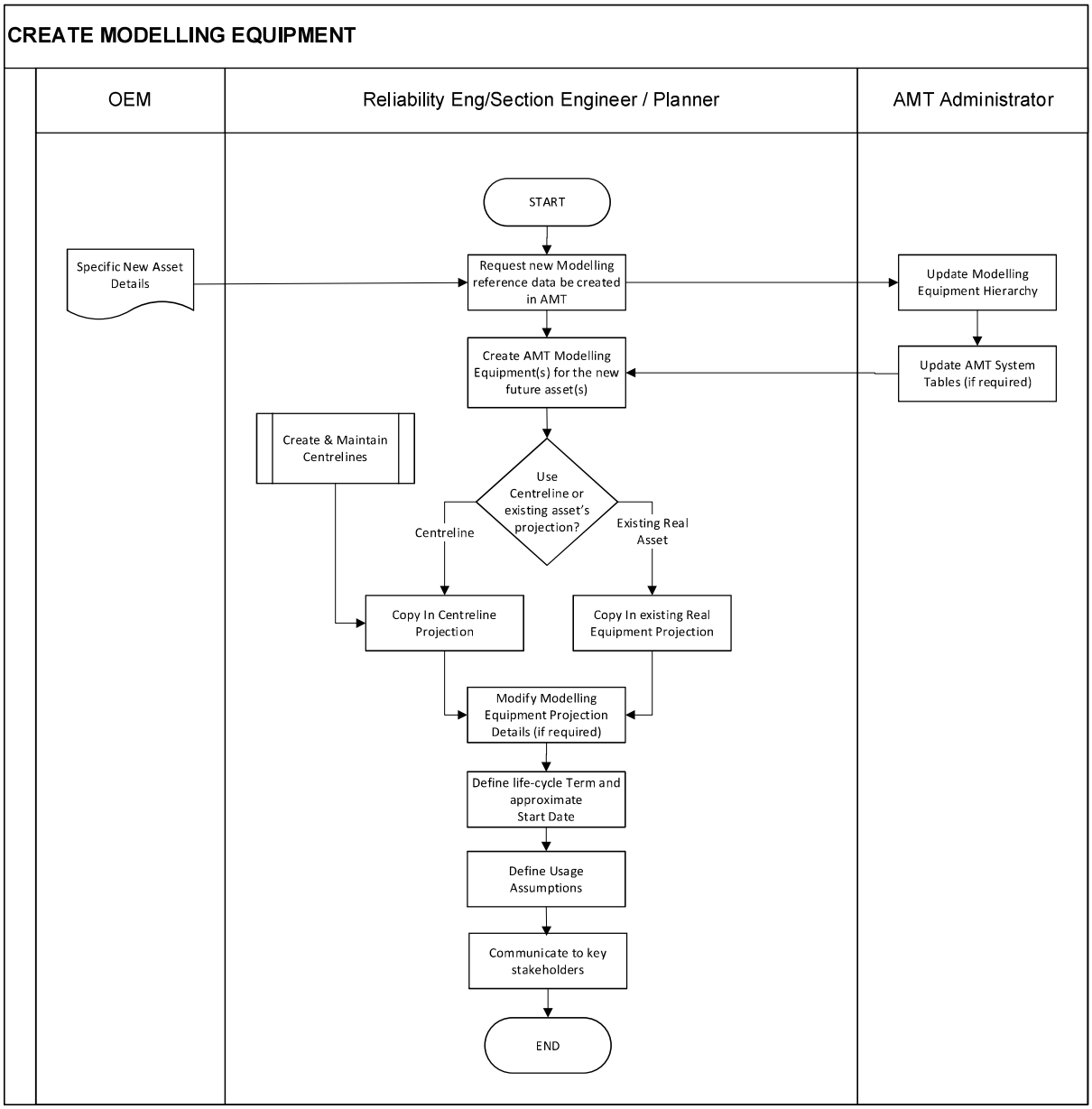
Modelling assets are fictitious AMT equipment, which represent future asset purchases. Modelling equipment are not real assets operating at MOPANI yet.

AMT can model and compare the direct maintenance costs of new assets between different suppliers/OEMs which MOPANI is considering purchasing.

The objectives of the process are:

Objective	Importance/Benefit
To create Modelling equipment in AMT which represent new equipment purchase	If new assets will be purchased by MOPANI, these future assets must be included in the maintenance budget and/or a maintenance reforecast. Modelling equipment are created in AMT for this purpose.

Figure 17 Modelling Assets Process Flow Chart



Process steps performed using AMT

Process Flow Step	AMT Function/Report	Notes
Update Modelling Equipment Hierarchy	Branch Administration Site Administration Fleet Administration	
Update AMT System Tables	System Tables > Maintenance	
Create AMT Modelling Equipment(s) for the new future asset(s)	Copy and Create Equipment - Modelling	

Process Flow Step	AMT Function/Report	Notes
Define life-cycle Term and approximate Start Date	Equipment Editor > Term	
Define Usage Assumptions	Equipment Editor > Term Usage Profile Manager	Usage Profile Manager only if Centrelines is configured for Advanced utilisation
Copy Strategy Tasks from an existing asset	Copy and Create Equipment - Modelling	
Copy In Centreline Projection	Copy and Create Equipment - Modelling	
Modify Modelling Equipment Projection Details (if required)	Equipment Editor > Tasks Multi-Task Editors	

The following outputs are expected from this process:

Output	Notes
Modelling equipment(s) created which can be imported into a budget and/or reforecast	

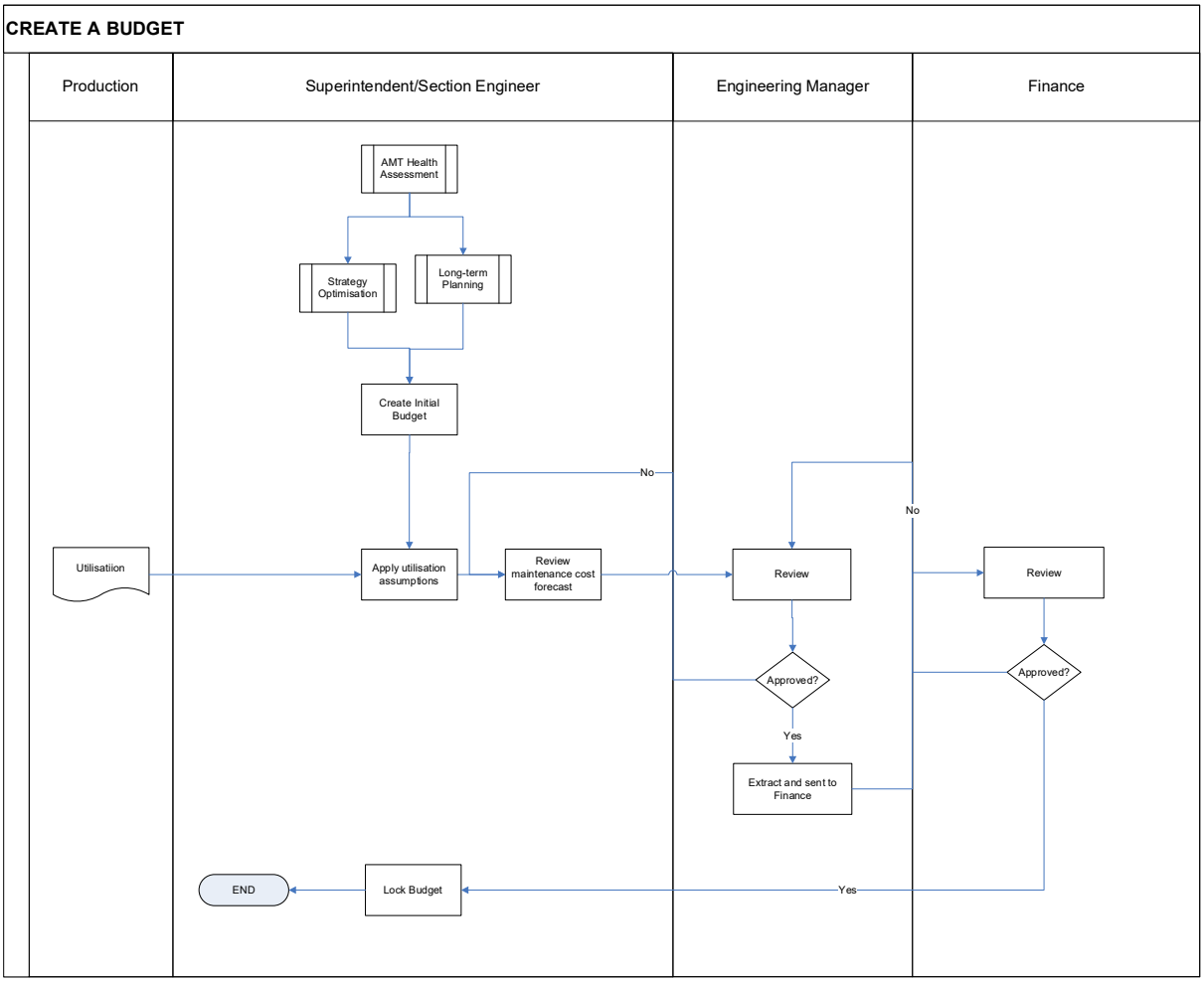
2.17. Create a Budget

AMT is to produce direct asset cost and maintenance budgets.

The objectives of the process are:

Objective	Importance/Benefit
Quickly and easily develop an accurate, zero-based asset budget for the asset management department.	Reduce time to build a budget. Budget forecast based on live LCC forecast.

Figure 18 Create a Budget Process Flow Chart



Process steps performed using AMT

Process Flow Step	AMT Function/Report	Notes
Create initial budget in AMT	Budget and Forecast Manager Budget and Forecast Manager > Equipment	
Apply utilisation assumptions	Budget and Forecast Manager > Usages	Copy/Paste new data with Excel
Adjust maintenance costs as required	Budget and Forecast Manager > Equipment Costs Budget and Forecast Manager > Cost Escalator	
Add new equipment to the budget (If applicable)	Budget and Forecast Manager > Equipment	Modelling equipment
Review the Budget and Approve		
Submit to relevant parties for Approval	Budget and Forecast Manager > Budget Reports	
Lock AMT Budget	Budget and Forecast Manager > Status = Locked	Complete Person Responsible, Scenario and Version fields

The following outputs are expected from this process:

Output	Notes
Direct equipment cost budget from AMT	
Locked and approved budget in AMT	Used to measure Actuals vs. Budget

Key Solution decisions regarding the **Budgeting** process:

- AMT will produce budget for direct asset maintenance costs.
- AMT will not be used to budget maintenance overhead costs.

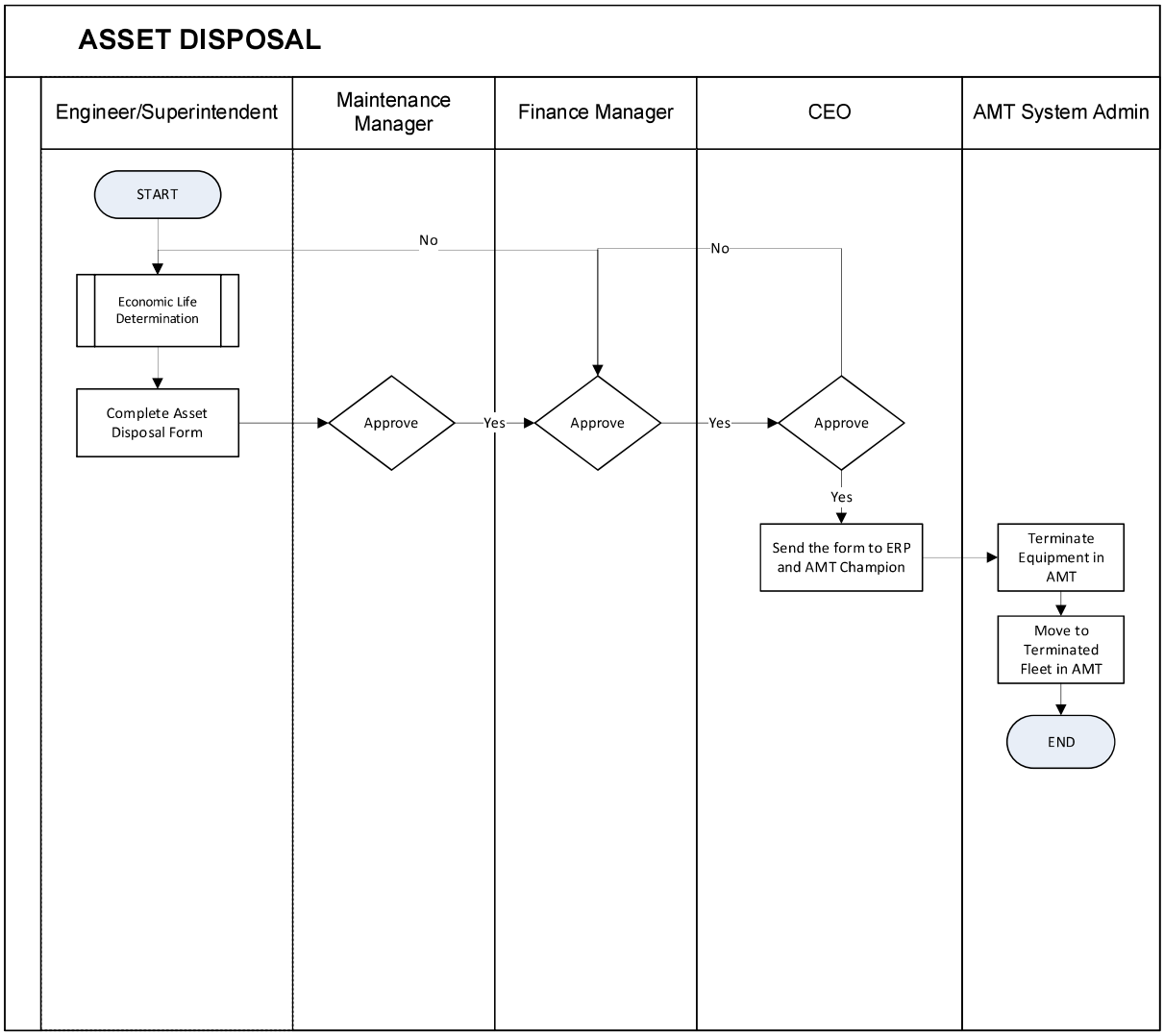
2.18. Asset Disposal

As asset is classified as "disposed" if it effectively no longer exists in the company, and therefore cannot be assigned to a project today or in the future. For example, an asset may be sold to another company, or it may be permanently inoperable. If an asset is classified as disposed, then in AMT the asset must be terminated. Note, even if an asset is terminated in AMT, its maintenance strategy and maintenance history remain intact and visible in the system for reporting and analysis purposes.

The objectives of the process are:

Objective	Importance/Benefit
Remove asset from the site	User can no longer manage the asset in AMT.
Inform AMT that an asset is terminated	AMT ceases forecasting all asset costs, components, labour, etc.

Figure 19 Asset Disposal Process Flow Chart



Process steps performed using AMT:

Process Flow Step	AMT Function/Report	Notes
Terminate Equipment in AMT	Equipment Editor > Term	The user may be asked to create an actual usage reading (i.e. machine hour reading) for the date the asset is deemed to be terminated. If so, use the Actual Usage Reading function to create the actual usage reading record on the same date the asset is terminated (i.e. same as the asset's End Date).
Move to Terminated Fleet	Equipment Editor > Base Details	Select the appropriate fleet in which all disposed assets are allocated.

Outputs from this process are:

Output	Description
Asset is terminated in AMT	Forecasting is ceased for the asset in AMT
Asset is removed from the site	Site users can no longer manage the asset (e.g. plan and scheduled work orders)
Asset is moved to a Disposed/Terminated Asset Fleet	Asset still exists in the system for reporting and analysis; for users which have access to this fleet.

3. AMT Functionality

The table below lists the AMT functionality that will be deployed.

AMT Functionality	Description
DLCC & Budgeting (Dynamic Life-Cycle Costing)	• Life-Cycle Costing and analytics • Forecasting and tracking of equipment maintenance costs • Strategy definition (including standard jobs) • Strategy optimisation • What-If analysis • Budgeting & forecasting • Long-term planning • Asset benchmarking • Asset economic life optimisation
Equipment Management (EM) Issues	• Quantify product problems (financial cost, productivity cost) • Workflow for actioning product problem resolutions
Planning and Scheduling	• Backlog management • Work order planning • Work order scheduling • Equipment Status (Shiftlog) • Downtime Capture • Equipment Performance KPI reporting • Maintenance Efficiency KPI reporting
Asset Health	• Condition Based Maintenance
Other Functionality	• Dashboard • AMT-IC custom report writing • Component Based Depreciation • Multi-currency

3.1. AMT Software Enhancements

- None identified.

3.2. AMT System-level Configuration

AMT has many global configuration settings which apply to all AMT branches, sites, fleets and assets, and AMT functionality, security, and integration. The following AMT system-level configuration decisions are summarised below.

Item	Description	Setting
EndUsageWarningLimit	Difference between the Current Date and Calculated Equipment End Date expressed in years (Warns the user if the difference is greater than the specified limit)	25
EndUsageErrorLimit	Difference between the Current Date and Calculated Equipment End Date expressed in years. (returns Error if the difference is greater than the specified limit)	25
DefaultAnnualUtilisation	If there is no annual utilisation provided to AMT by ERP then the asset is created in AMT as a usage-based asset with this annual utilisation rate, and using the system default UOM. Note: the default UOM for MOPANI's AMT = Hours.	8800
DefaultEquipmentEndDate	If no End Date or End Usage is provided to AMT by the ERP then the asset is created in AMT with this end date. Note: assets will not be imported into AMT but setting still applicable for bulk loading data through load sheets	120 000
Asset Register	Enable the ability to capture asset estimated replacement cost, estimated disposal price against an AMT asset.	No
LoadEqpExplorerAtStartup	Automatically load AMT's equipment hierarchy explorer at start up.	Yes
AutoUpdateUtilisation	The system task runs for simple utilisation asset with auto update annual utilisation flag switched on. If Average utilisation for last (AutoUpdateUtilisationMonths) months varies by > (AutoUpdateUtilisationMarginPct)% from set utilisation then it will reset the utilisation to the average for the last (AutoUpdateUtilisationMonths) months.	Enabled
AutoUpdateUtilisationMonths	Period in months to calculate Average Utilisation.	6
AutoUpdateUtilisationMarginPct	Margin in percent between annual utilisation set for equipment and average utilisation for the last number of months (number of months is set in AutoUpdateUtilisationMonths).	10%
PlanningDaysDefault	If no value is provided for planning days, then the projected task is created with this number of planning days.	14
Strategy_Reschedule	Yes = Use the "strategy reset" functionality (i.e. this will reschedule unplanned strategy tasks when they become overdue. This prevents the "bow-wave"), No = Do not use the "strategy reset" functionality.	Yes
Pct_Frequency_Reschedule	If Strategy_Reschedule = Yes, this is the percentage of the frequency by which overdue occurrences of unplanned strategy tasks are moved into the future.	33
DefaultToleranceHours	When new Suppression Groups are created by ERP, this is the default tolerance hours.	48

Item	Description	Setting
DefaultProjTaskJobCodeId	If no job code is provided by ERP, then the AMT Strategy Task is created with this job code.	00
AttachedDocumentCleanup Delay	A system task deletes documents that exist on the server that are no longer attached to records in the AMT database. This setting is the number of days before such "rogue" documents are deleted.	90
DaysUsageToOverwrite	Usage readings can be overwritten for this number of days before the most recent actual usage reading is imported.	14
Thimalini	This time limit (in days) is used in some Quick Watch triggers (e.g. Breakdown and First Stoppage) to provide only recent records.	14
UsageTolerance	Equipment actual usage tolerance (%).	500
NextOccurenceDate Logging	Enforce logging of comments after updating the Next Occurrence Date in Long-term Planning.	True
Max_Comp_Life	Default value for percentage of Maximum Component Life Used (High Risk/High Opportunity reports).	20
Min_Comp_Cost	Default Minimum Component Cost (High Risk/High Opportunity reports).	10,000
Delete User Amt Log	This AMT system task deletes data from log tables. By default, the system task deletes entries for user logins which are 1 month old. The system task has parameters: EntityNameTable - for log type to delete; and MonthDecrement - the log entries which are older than current date - MonthDecrement months will be deleted. The system task can be configured to delete any log type in the any date range.	2 (Prefer 90 days)
MaxAttachmentSizeInKb	The largest size in KB of an attachment that can be uploaded by the AMT Client application.	8000 (Check maximum allowed in AMT)
NoOfPeriodsToMerge	Applies to merging rule 2. Merging is based on Eqp, WO# and dates within No of Periods to Merge months. Note: It should be a high number because we want all WOs with the same WO Number to merge. Suggesting 99 as there should not be a WO spreading over 99 financial Periods.	99
Currency	All reporting will be in AED, and all transactions imported into AMT will be in AED.	AED
Delete_Missing_Segments	When importing a Work Order, AMT will delete all segments and expect all segments to be provided in the Work Order Import file. We do not want to delete the missing segments as we are bringing the costs using the Settlements rather than the usual way of using the WO Details.	Yes
AutoCreateWOCheckedDefault	AMT will be configured in auto-create work order (ACW) mode.	True

Item	Description	Setting
ShowAutoCreateWO	AMT will be configured in auto-create work order (ACW) mode for all Branches (not optional).	True
ACWManualWOClosure	No work order financial closure process for auto-create work orders (auto set to closed).	False
ACWTaskDeletable	Auto-create work orders can be deleted by users.	True
WOUniqueIdentifier	Auto-create work orders will be prefixed with a character.	"W"
ShowPOActions	Enable ability to email print Purchase Orders / Requisitions.	Disabled

3.3. AMT User Security-level Configuration

3.3.1. Groups

- Groups will be set up in AMT per User Role, with specific permissions
- User Roles will be Site specific
- Group and group permissions will be administered by the AMT Champion

3.3.2. User Accounts

- User Accounts will be administered by the AMT Champion
- Users will be associated with a:
    - Group
    - Language
    - Branch / Site / Equipment access

3.4. AMT System Table Configuration

AMT contains several system or reference tables. As part of the AMT deployment at MOPANI these system tables need to be defined by MOPANI before master and transactional data can collected (since this data references system tables). The system tables need to be a standard for all MOPANI sites. Many system tables will map directly to reference data tables in ERP.

Below is the list of AMT system tables for the current release of AMT, as of the date of this document, which are relevant for the Solution. If an AMT system table is based on ERP reference data, then this is indicated with "Yes" in the "Mapped to ERP" column and the ERP reference table is shown in the "ERP Source" column.

System Table Grouping	System Table Entity	Mapped to ERP	ERP Source
Organisation	Fleet	No	
Financial	Cost Bearer	Yes	Account Code

System Table Grouping	System Table Entity	Mapped to ERP	ERP Source
Financial	Cost Centre	Yes	GL Code, Activity code
Maintenance	Component Code Hierarchy	No	MOPANI will use own defined codes
Maintenance	Modifier Code	No	MOPANI will use own defined codes
Maintenance	Task Type	No	MOPANI will use own defined codes
Maintenance	Task Counter	No	MOPANI will use own defined codes
Maintenance	Job Code	No	MOPANI will use own defined codes
Equipment	Manufacturer	Yes	ERP equipment table
Equipment	Model	Yes	ERP equipment table

3.5. AMT System Data Entities

The main data entities and a description of how each will be maintained are detailed below:

Item	Description
Users	Manually maintained in AMT
User Roles	Manually maintained in AMT
Assets	Manually maintained in AMT
Actual Usages	Manually maintained in AMT and will be integrated in future
Usage Profiles	Manually maintained in AMT
Strategy Tasks	Manually maintained in AMT
Strategy Tasks – Document links	Manually maintained in AMT
Work Group Available Labour Hours	Manually maintained in AMT
Work Order Settlements	Work order settlements will be imported into AMT via integration
Actual Downtime	Manually maintained in AMT

3.6. AMT IT Architecture Configuration

- AMT will be installed on two different MOPANI environments - Test and Production.
- EPF will be installed on its own server environment.
- The AMT Production environment should be on a single dedicated application server and separate database server (both of which may be virtual machines, but neither should use host hardware shared with other applications).
- The AMT Test environment should be on a separate server (which may be a virtual machine) linked to a common database server.
- AMT Production and Test servers should meet minimum hardware requirements for AMT as documented in RPM's "AMT - Hardware & Platform Requirements" document.
- AMT users will be located on MOPANI sites with reasonable network access.

- MOPANI's AMT users will access AMT servers using the internal company network or over a secure internet connection.

4. Deployment Scope

The following sections outlines the key aspects of deploying AMT for MOPANI.

4.1. Sites and Equipment

Site	Location	Asset Category
Mufulira	Zambia	Mobile and Fixed Plant
Nkana	Zambia	Mobile and Fixed Plant

4.2. Data Preparation

- The following data will be required for the new AMT solution
    - System Tables
        - AMT has several system tables that need populating which fall into 4 categories – Organisation, Financial, Maintenance and Equipment.
    - Asset Register
        - Equipment hierarchy
        - Active equipment
        - Current projection strategy tasks
        - Budget projection strategy tasks (Only for new future purchased assets.)
        - Strategy Task Next Occurrences
    - Transaction History
        - Historical work orders for the past 2 years will be loaded into AMT.
    - Current backlogs in open state
    - Usage History
- All data will be collected by MOPANI in templates provided by RPM.
- RPM will guide MOPANI on the templates, and review progress.
- MOPANI will collect data for all sites in the templates provided by RPM (not a phased approach)
- RPM will load templates into AMT. Once loaded, there will be a single Data Cleansing workshop. If more data cleansing is required, MOPANI will continue to cleanse the data independently.
- RPM has budgeted single rounds of data loading by RPM into MOPANI's AMT Test and Production environment, respectively. Each Phase (1 AND 2) will have a single load of data

4.3. User Training

- RPM will follow a Train-the-Trainer preferably should be AMT champions for each site
- RPM will deliver a single classroom training course to MOPANI's nominated AMT trainees
- MOPANI AMT Champions would then train all AMT end users for go live.
- RPM will deliver additional system administration training for MOPANI's AMT Champions and Mopani IT System Administrators
- MOPANI will nominate at least one AMT Champion who will provide ongoing training and 1st level support as required for AMT users.

4.4. Change Management

- MOPANI is responsible for any business process re-engineering activities and associated change management activities to redesign and/or implement new or enhanced business processes and procedures in preparation to roll out of the standard AMT solution.

4.5. Rollout Approach

- RPM is responsible to support all sites pre- and post-go live for 2 weeks.

4.6. MOPANI Project Team (Refer to Project Governance)

- MOPANI will establish a dedicated, central project team who will be responsible for the AMT implementation project across the organisation.
- A MOPANI resource who is part of the central project team will have authority to issue directives and make key decisions with respect to AMT configuration settings, data, and system setup.
- There will be a central project manager, AMT champion(s), and system administrators.

4.7. Data Transfers between ERP and AMT

The table below lists the data entities which will be transferred between ERP and AMT.

Interface	Direction	Description
Part, Parts Prices	ERP to AMT	Defines the messages used to: • import new Parts into AMT • update Part Prices for Parts already registered in AMT
Inventory Parts	ERP to AMT	Defines the messages used to import Inventory Parts into AMT from ERP System.
Supersede Parts	ERP to AMT	Defines the messages used to import Superseded Parts information into AMT from an ERP System.
Inventory Search & Validation	AMT to ERP	Defines the messages sent from AMT to an ERP system to perform the Inventory Search and Inventory Validation functions in the AMT Work Order.

Interface	Direction	Description
Create Purchase Order	AMT to ERP	Defines the messages sent from AMT to automate the creation (and update) of Purchase Orders in an ERP/Purchasing system.
Work Order Details (Part Order Status)	AMT to ERP	Defines the messages used to update Parts status in AMT from an ERP system.
Work Order Details (Actuals)	ERP to AMT	Defines the messages used to update workorder actuals in AMT from an ERP system.
Work Order Financial Summary	ERP to AMT	Defines the messages used to import and export Workorder Settlements between AMT and ERP systems.
General Ledger	ERP to AMT	Defines the messages used to import General Ledger data into AMT, to support the process for reconciling AMT costs with General Ledger costs in the ERP System.

Notes:

- The frequency with which data will be transferred between ERP and AMT has yet to be defined.
- On a periodic basis MOPANI will manually extract the data from ERP and then manually transfer the data to AMT.
- The data extracted by MOPANI from ERP will conform to specific rules and formats provided by RPM.
- Field mappings related to the data extraction from ERP are maintained by MOPANI.